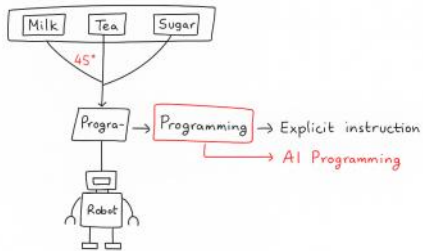


Feature	AI	ML	DL
Definition	AI aims to create machines that can perform human-like intelligent tasks.	ML is a subset of AI where machines learn from data automatically.	DL is a subset of ML that uses multi-layer neural networks to learn complex patterns.
Works On	Logic, reasoning, rules, decision-making	Data + algorithms	Large datasets + neural networks
Human Intervention	High (rule-based systems)	Medium (feature engineering required)	Low (auto feature extraction)
Data Requirement	Can work with small data	Needs moderate data	Needs huge data
Speed	Slower (due to symbolic reasoning)	Faster than AI	Fast during prediction but requires heavy training
Techniques Used	Search algorithms, expert systems, planning, NP	Regression, classification, clustering	CNN, RNN, Transformers, GANs
Hardware Requirement	Normal CPU	CPU	Mostly GPU/TPU
Output Type	Decisions	Predictions	Highly accurate predictions
Scope	Broadest	Narrower	Most specialized

What is an AI?

Task → Go to Kitchen prepare tea



→ SML →
← UML ←

ML

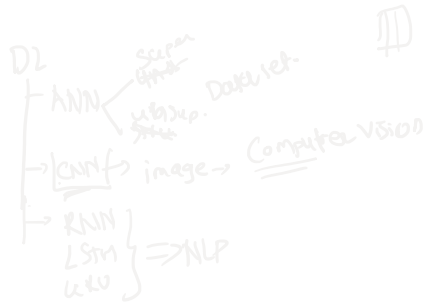
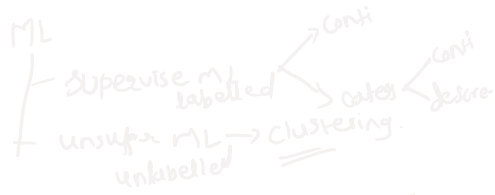
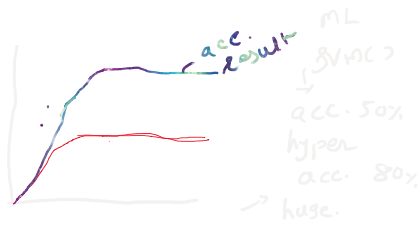
DL

- 1) Structured.
- 2) limited ad
- 3)
- 1) unstru,
- 2) large dataset
- 3) train model again & again to min. error.

acc. result
ML
SVM

FD

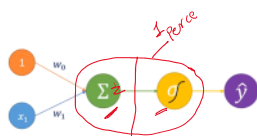
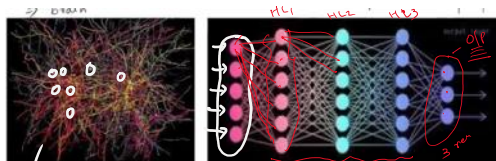
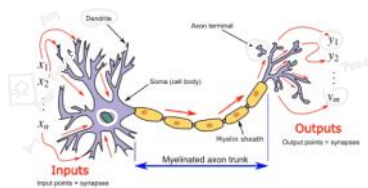
15h



AIA → a/n

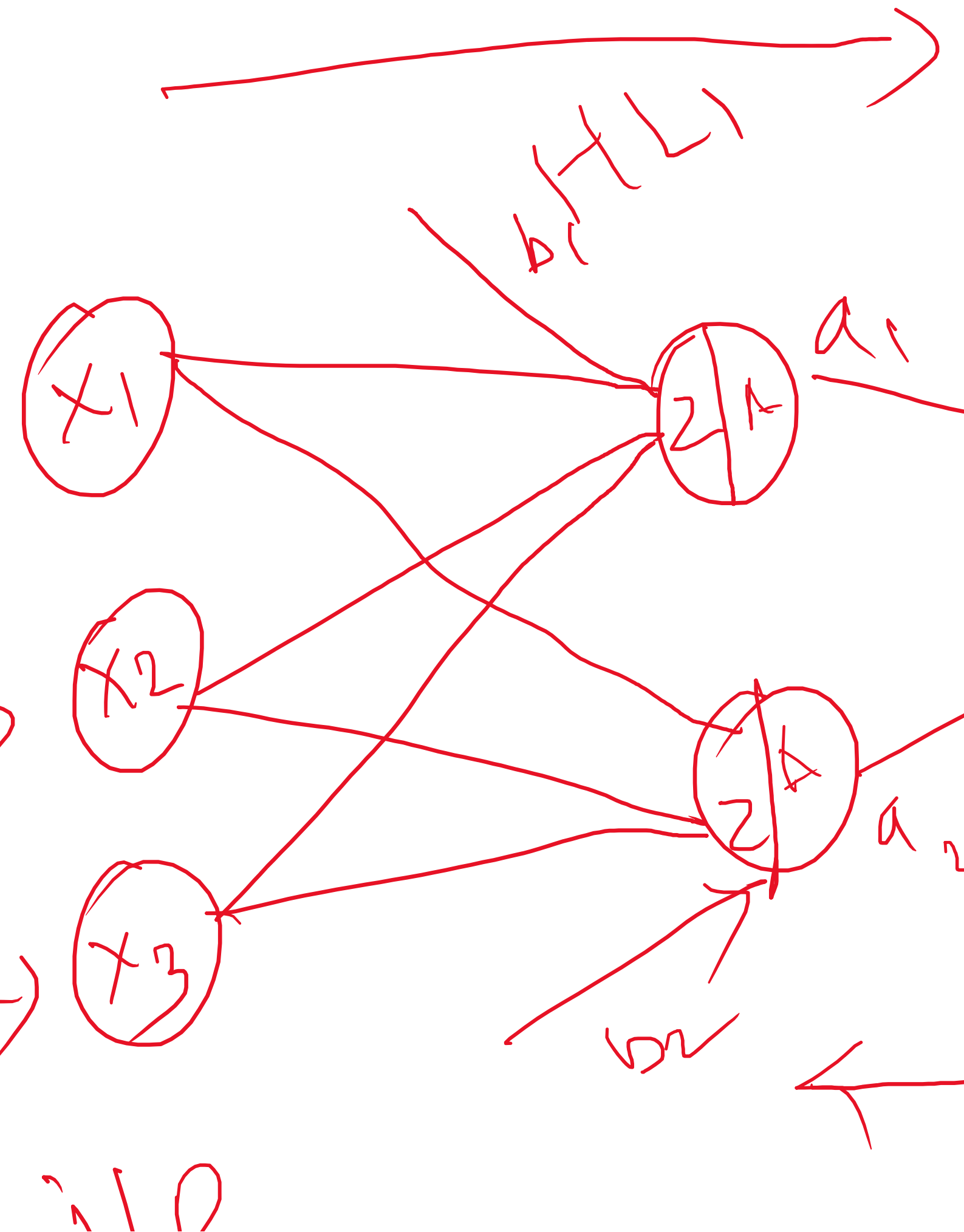
CVE →

NLP →



- 1) Perceptron
- 2) MLP
- 3) FP / Forward Propagation
- 4) need of Activation fun.
- 5) math - derivatives ✓
- 6) BP / Backward Prop.
- 7) chain of rule ✓
- 8) Types of activation function
- 9) losses
- 10) weight initialization tech.





1-X

0.2

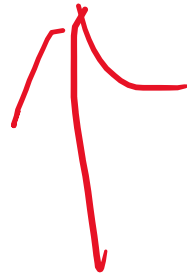


O/p

BP - Optimizers

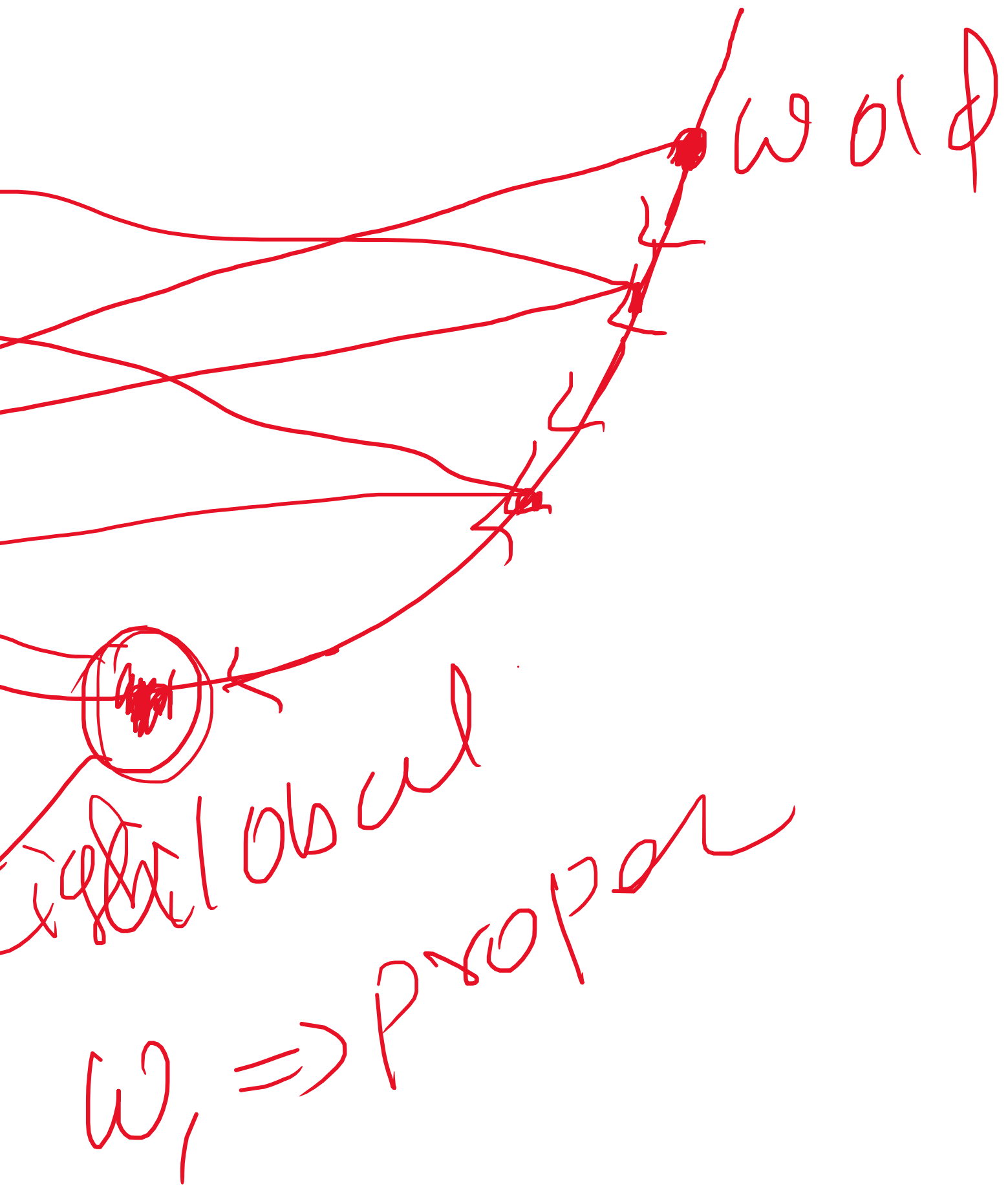
Error

$$= y_A - y$$





we



Given:

- $x = 2$
- $y = 10$
- $w = 3$
- $\alpha = 0.1$

Prediction:

Loss:

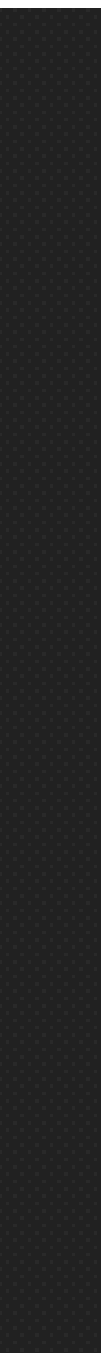
Derivative:

$$= \omega \cdot x \cdot b \quad b=0$$

$$\hat{y} = 3 \times 2 = \underline{6}$$

$$= (\underline{10 - 6})^2 = \underline{16}$$

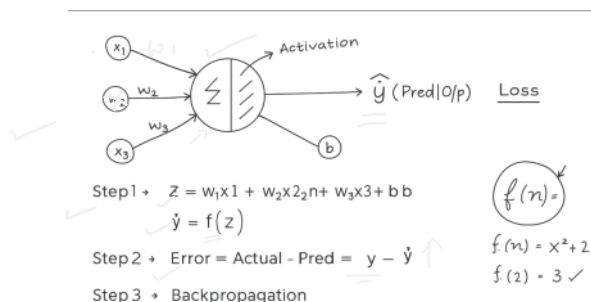
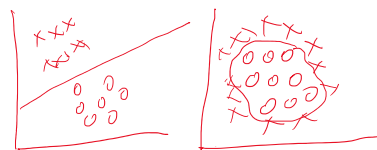
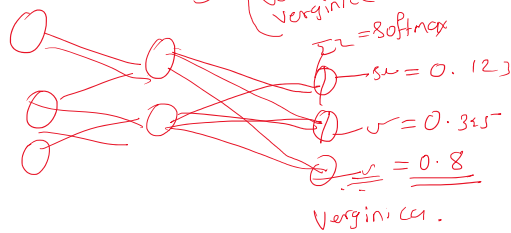
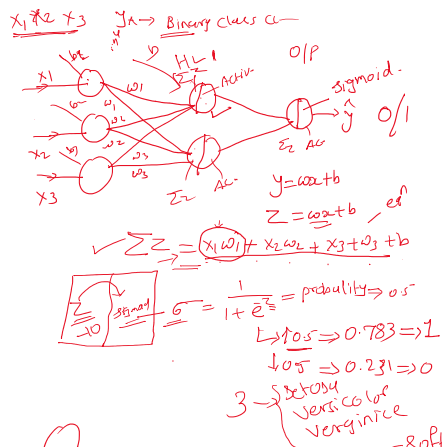
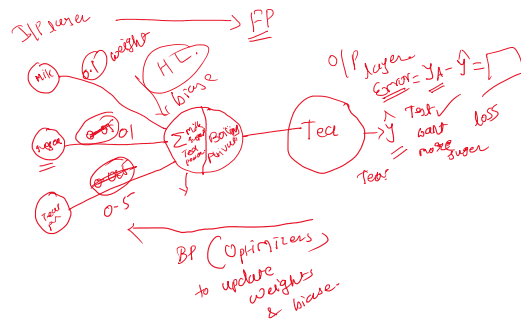
$$\begin{array}{r} 3 \\ 4-6 \\ \hline \hline \end{array}$$







- 9) loss
- 10) weight initialization tech.
- 11) Optimizers (gradient descent)
- 12) Batch normalization.



VP

mayor

DI - UP

GRADIENT

How Gradient Descent works

Gradient descent is an optimization algorithm that minimizes the error by updating the weights. Steps:

- Randomly start with weights and find optimal weight till the error is less.
- It uses the concept of partial derivatives to update weights till it reaches global minimum

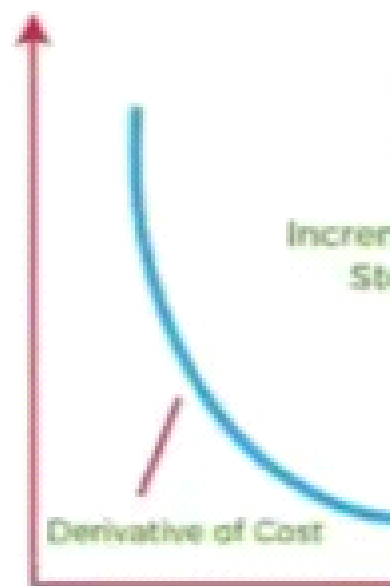
$$w_{new} = w_{old} - \eta \cdot \frac{\partial L}{\partial w}$$

$$b_{new} = b_{old} - \eta \cdot \frac{\partial L}{\partial b}$$

DESCENT

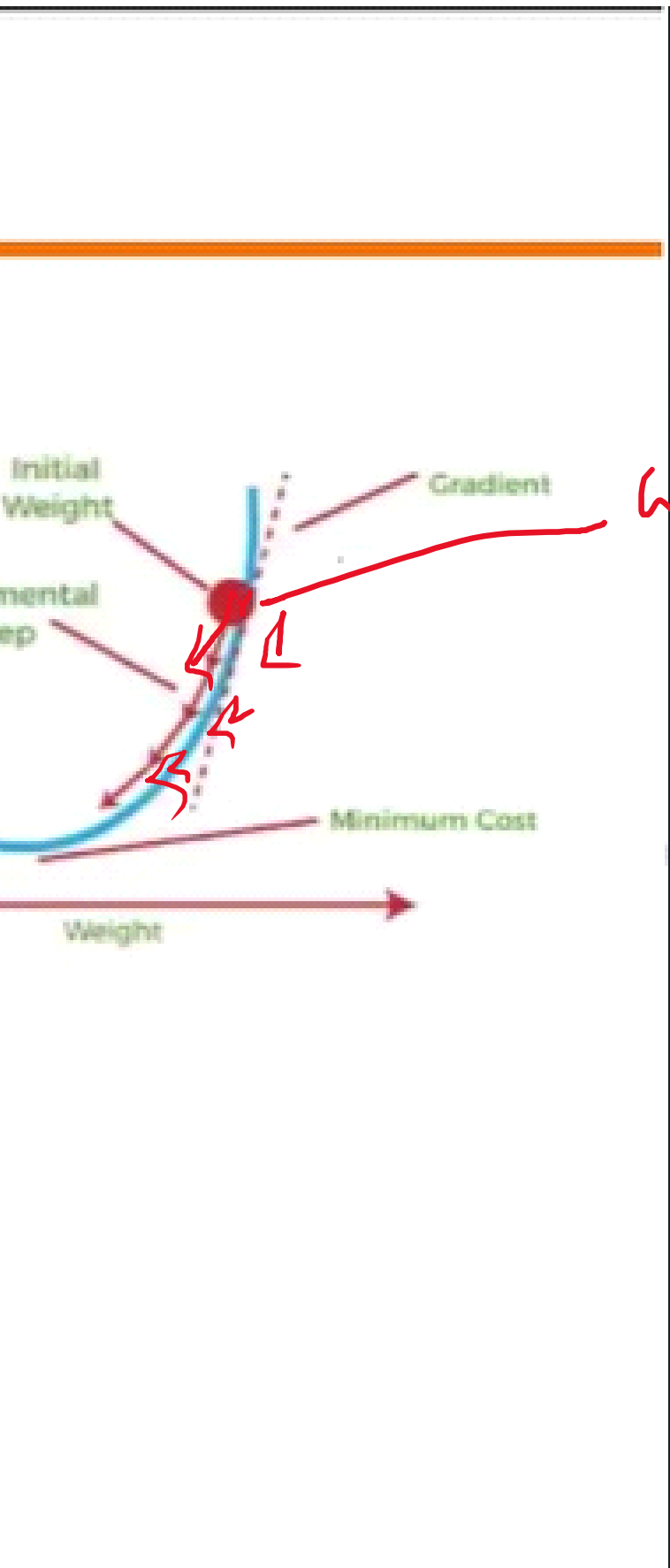
s?
izer which is used to minimize
ights.

ghts and works iteratively to
reaches global minima where
partial derivatives $\oplus$ update the
bal minima i.e where loss is



$$l_d - \alpha \frac{\partial L}{\partial w}$$
$$l_d - \alpha \frac{\partial L}{\partial b}$$

w
 $L \uparrow \downarrow$
 d



Derivative:

$$\frac{dL}{dw} = -$$

Weight Update:

$$w_{\text{new}} =$$

$$\underline{w} = 3$$

✓ New weight is closer to the correct value

✓ What is Chain Rule? (1-Line)

Chain Rule is used when a value depends on another value.

In ML:

nginx

Loss → Prediction → Weight

So we must use chain rule.

1 Model Setup (Same as Before)

Model:

$$-2(2)(10 - 6) = -16$$

$$= 100 - 2 \frac{dL}{d\omega}$$

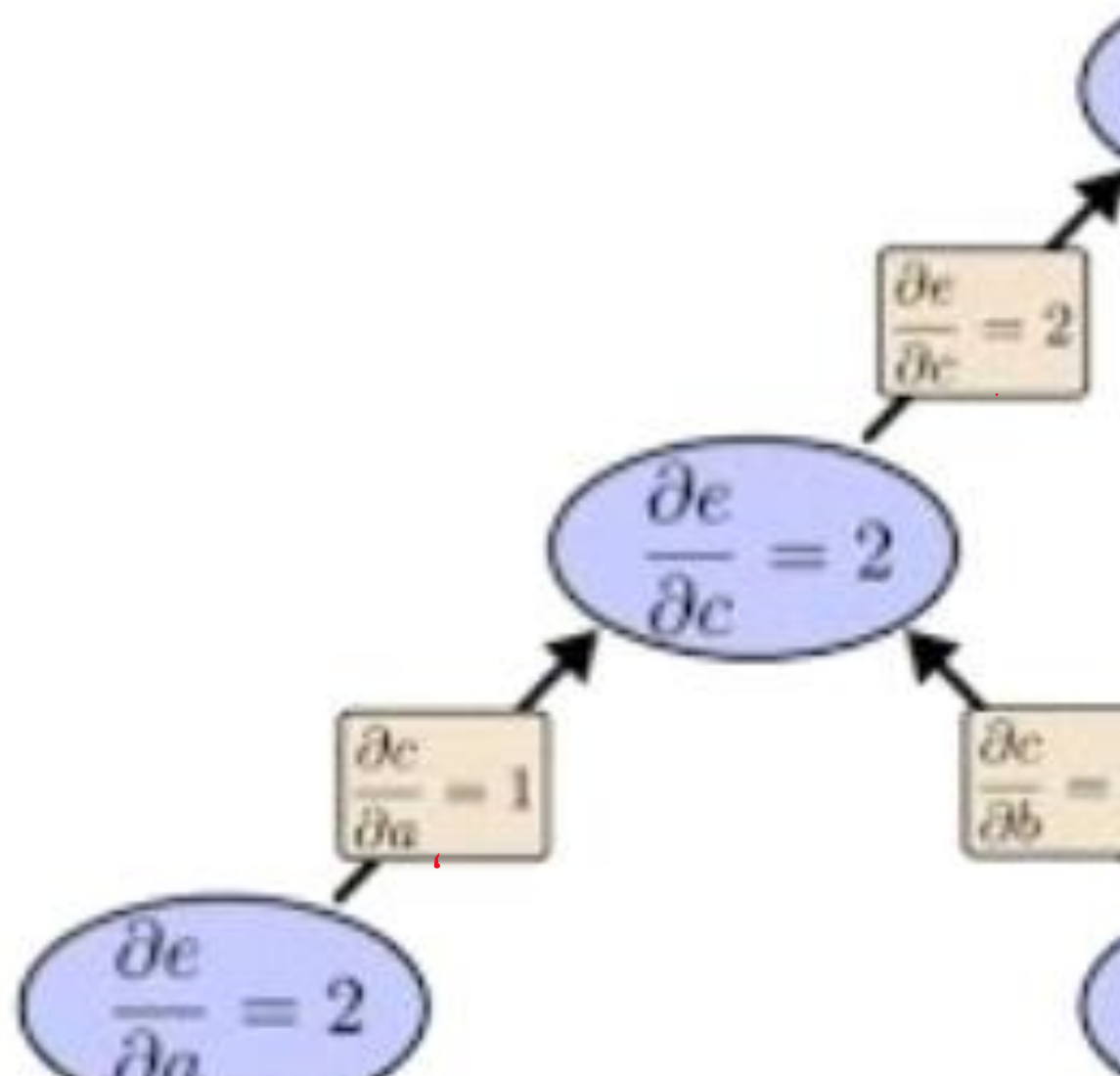
$$3 - 0.1(-16) = 4.6$$

, which again depends on another value.

 Copy code

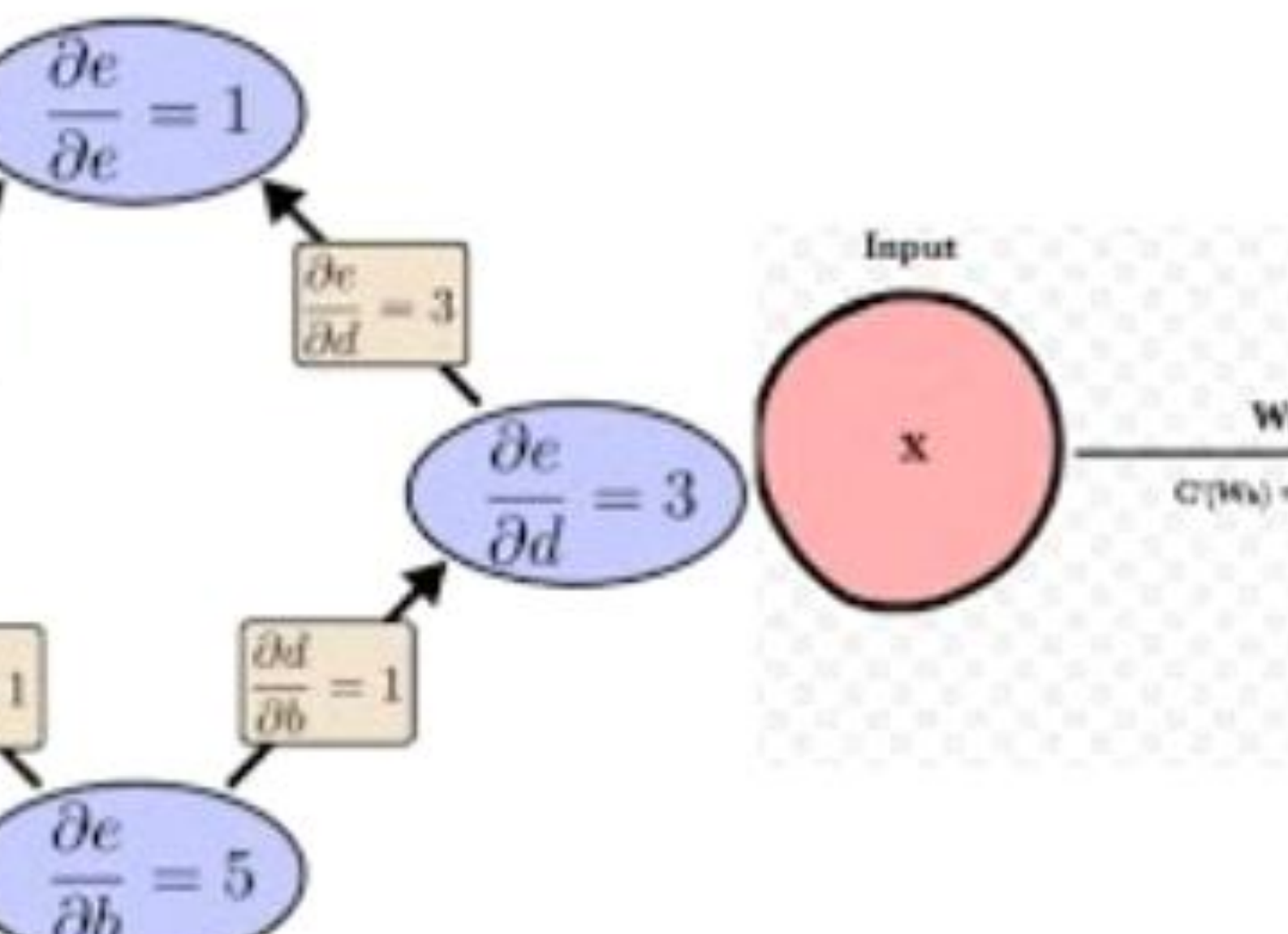
Deep Learning

Chain Rule



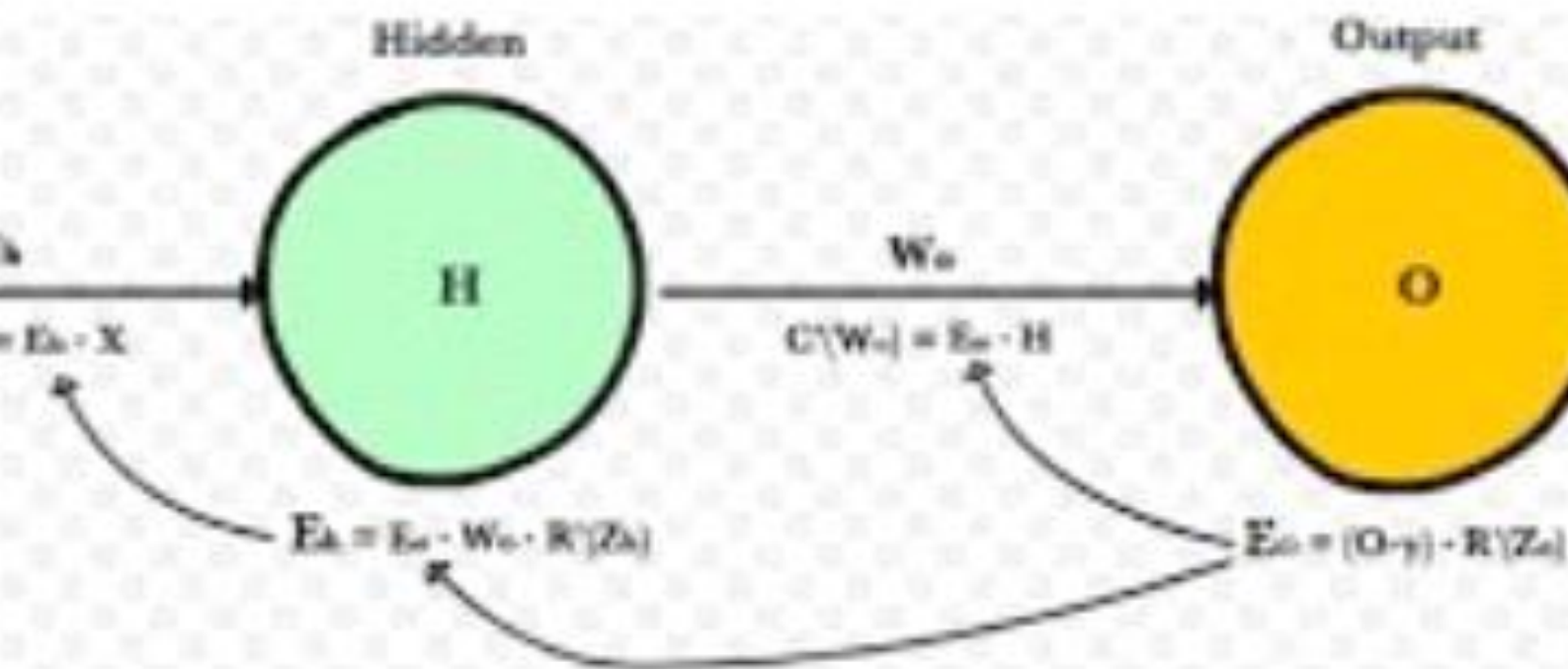
Learning Back Prop

Rule of Differ



agation

entiation





u.

u2

$$\hat{y} = f(z)$$

Step 2 • Error = Actual - Pred = $y - \hat{y}$ ↑

Step 3 • Backpropagation
Weights are adjusted

$$f(n) = x^2 + 2$$

$$f(2) = 3 \checkmark$$

MLP → ③ layer

① I/P

② H/L

③ O/P

H/L 1st neuron

$$\sum z_1' = x_1 w_{11}' + x_2 w_{12}' + x_3 w_{13}'$$

$$L=1 \quad f(z_1')$$

$$a_{11} = \text{relu}(z_1')$$

$$\sum z_2' = x_1 w_{12}' + x_2 w_{22}' + \dots$$

$$L=1 \quad f(z_2')$$

$$a_{12} = \text{relu}(z_2')$$

O/P layer → Binary

$$\sum z_1^2 = a_{11} w_{11}^2 + a_{12} w_{12}^2$$

$$\hat{y} = \sigma(z_1^2)$$

$$\hat{y} = 0.586$$

$$\hat{y} = 0.1$$

$$\text{Error} = y_A - \hat{y}$$

Step-1 → FP

$$\text{I/P} \rightarrow \hat{y}$$

$$\sum z / \text{act.}$$

Step-2 → loss

Reg → MSE, MAE, RMSE

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

... calculation

$$z_1 + b_1$$

$$x_3 \omega_{23}' + b_2'$$

class classif.

$$z_2 + b_1$$

$$\Rightarrow 1$$

$$0$$

—

Differentiation is a method of finding the rate of change of one quantity with respect to another quantity.

Formal Definition

If a quantity y depends on a quantity x , then

Then differentiation of y with respect to x is denoted by

which represents the rate of change of y with respect to x .

α

mathematical process used to find how fast one quantity changes w

ion

ds on another quantity x :

$$y = f(x)$$

finds:

$$\frac{dy}{dx}$$

rate of change of y with respect to x .

with respect to

$\frac{dL}{du} = \frac{d}{du} \left(\frac{1}{2} (y - \hat{y})^2 \right)$

🔥 This is Chain Rule

📌 Step-by-Step Differentiation Using Chain Rule

✅ Step 1: Derivative of Loss w.r.t Prediction

$$\frac{dL}{d\hat{y}} = -(y - \hat{y})$$

✅ Step 2: Derivative of Prediction w.r.t Weight

$$\frac{d\hat{y}}{du} = 1$$

✅ Step 3: Apply Chain Rule

$$\begin{aligned} \frac{dL}{du} &= \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{du} \\ &= -(y - \hat{y}) \cdot 1 \\ &= -2(y - \hat{y}) \end{aligned}$$

✅ Same gradient as before, but now you understand why!

📌 General Chain Rule

If:

Then:

Step 1: Model (Prediction)

Suppose our model predicts $\hat{y}$ given input x

$$\frac{dL}{dy} \cdot \frac{dy}{du}$$

Chain Rule

$$2(y - \hat{y})$$

$$y - \hat{y} = x$$

$$\frac{dL}{dy} \cdot \frac{dy}{du}$$

$$2(y - \hat{y}) = x$$

why?

Chain Rule Formula (Exam-Ready)

$$L = f(g(w))$$

$$\frac{dL}{dw} = f'(g(w)) \cdot g'(w)$$

Numerical Example (With Chain Rule)

Given:

- $x = 2$
- $y = 10$
- $w = 3$

Forward Pass:

$$\hat{y} = 3 \times 2 = 6$$

Backward Pass (Chain Rule):

$$\frac{dL}{dy} = -2(10 - 6) = -8$$

$$\frac{dy}{dw} = 2$$

$$\frac{dL}{dw} = -8 \times 2 = -16$$

■ This gradient tells how to update w

(Prediction Function)

Model is:

$$\hat{u} = w \cdot x + b = 0$$

Model:

$$\hat{y} = w \cdot x$$

Loss Function:

$$L = (y - \hat{y})^2$$

Here:

- L depends on $\hat{y}$
- $\hat{y}$ depends on w



$$\frac{\partial L}{\partial a} = 2$$

$$\frac{dL}{dx} =$$

$$y = x^2 \quad \frac{d}{dx}(x^2)$$

$$\overline{\partial b} = 5$$

$$\frac{dL}{du} \sim \frac{dL}{da}$$

$$2x$$

2

$$\frac{dL}{dW} = \frac{d}{dW}$$



u1

u3

u5

dr



u4

du1



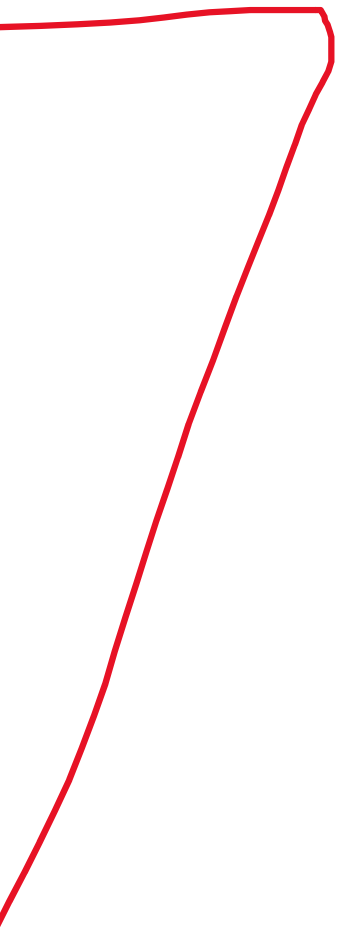
u1



du2

rain water

duz dus
da



~~N~~ ~~n~~

Binary Class Classification
→ Binary Cross Entropy / log loss

↳ log →

multi class classification

↳ Categorical Cross Entropy

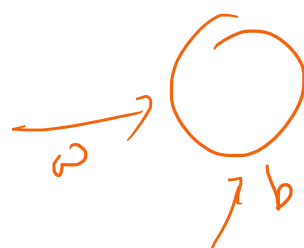
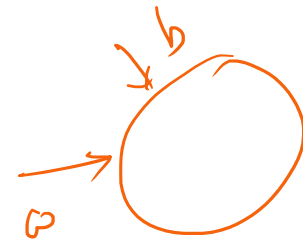
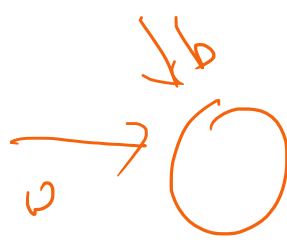
↳ Sparse Categorical cr



Step-3 BP → updating weights

(Optimizer →

1 epoch = $FP + Loss + BP$



ω
0,1

biase

$$y$$
$$z = x(\omega) = x \cdot 0 + b$$
$$= 0$$

weight

derivative

$$\omega_{new} = 0$$



vectors $\rightarrow$ direction

as well



MAGNITUDE AND DIRECTION OF A VECTOR

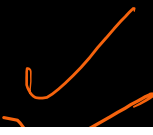


Find the magnitude and direction of each vector a b

$$\vec{v} = \langle a, b \rangle$$

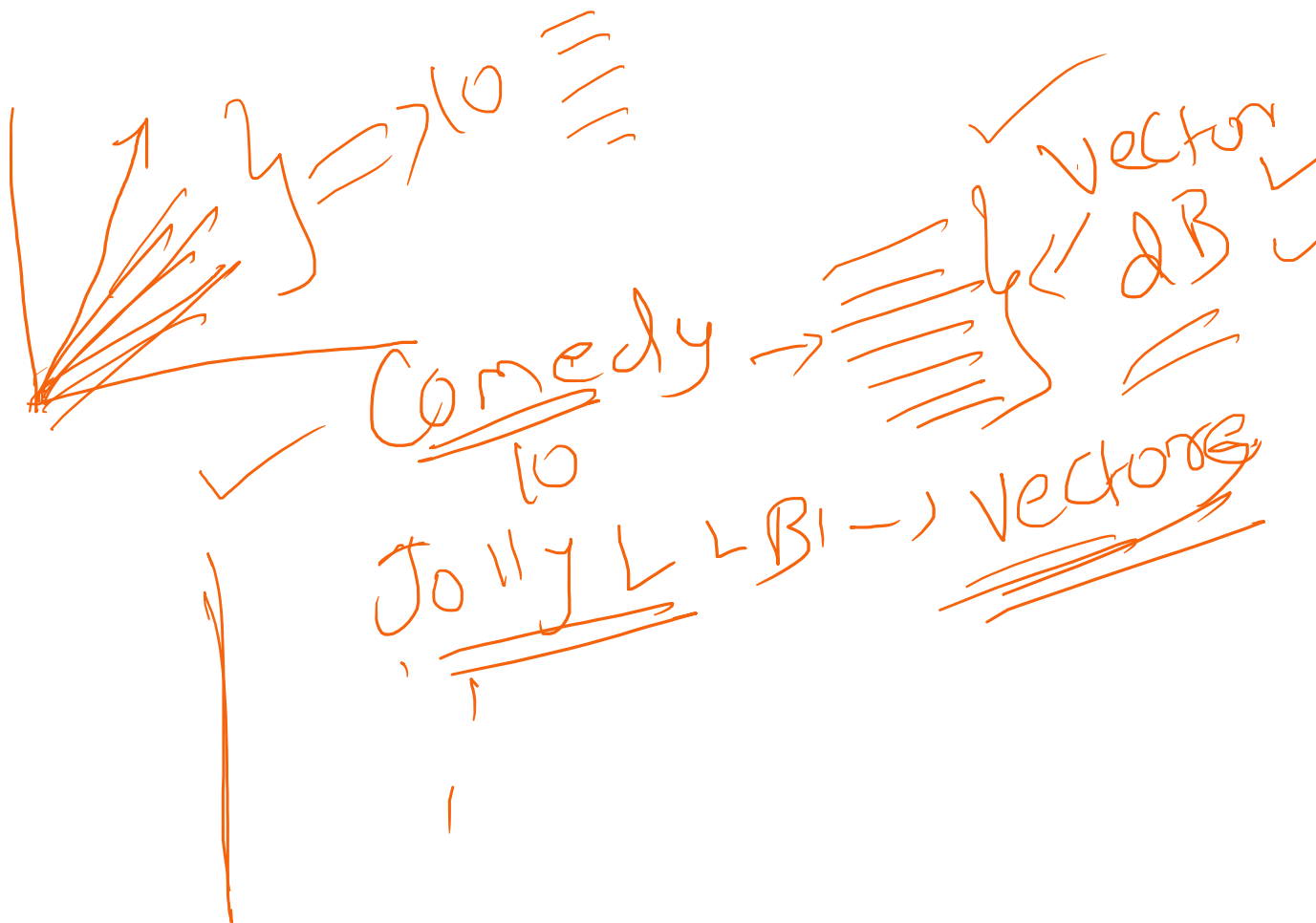
Magnitude $\|\vec{v}\|$

$$\|\vec{v}\| = \sqrt{a^2 + b^2}$$



ection

magnitude



Where:

- x = input feature
- w = weight
- $\hat{y}$ = predicted value

Step 2: Loss Function

We use Mean Squared Error (MSE)

This loss tells how far the predicted value is from the actual value

Step 3: Derivative



This derivative is used to update the weights

$$\hat{y} = \underline{w \cdot x} + \underline{b = 0}$$

ature

(parameter to learn)

d output

nction (Error)

uated Error (MSE):

$$L = (y - \hat{y})^2 = (y - wx)^2$$

w wrong the prediction is.

ifferentiate the Loss w.r.t. Weight

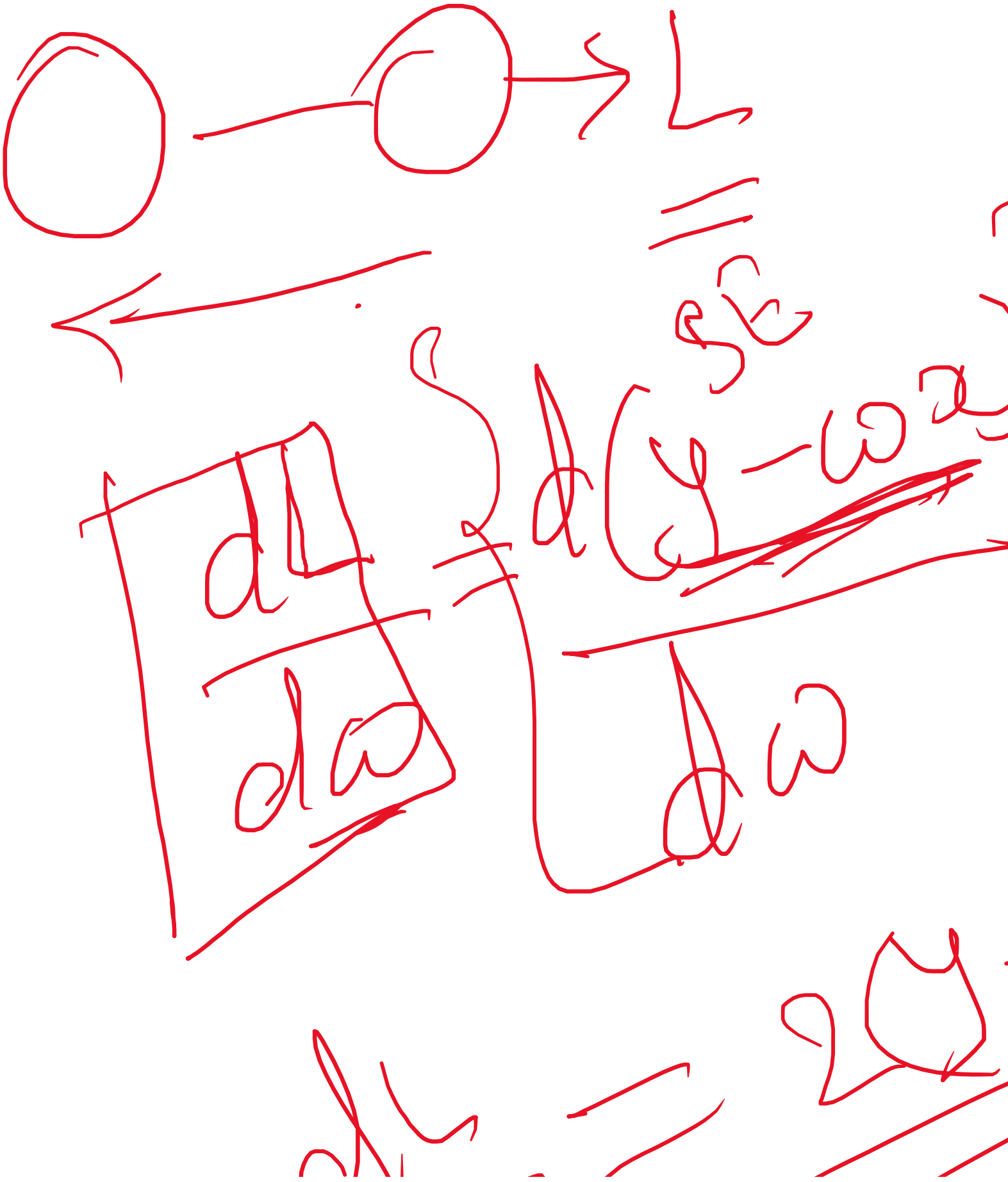
$$\frac{dL}{dw} = -2x(y - wx)$$

ivative tells:

0

0

$$\frac{d^2 r^2}{dr^2} = 2^3$$



$$\frac{du}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx}$$

✓

✓

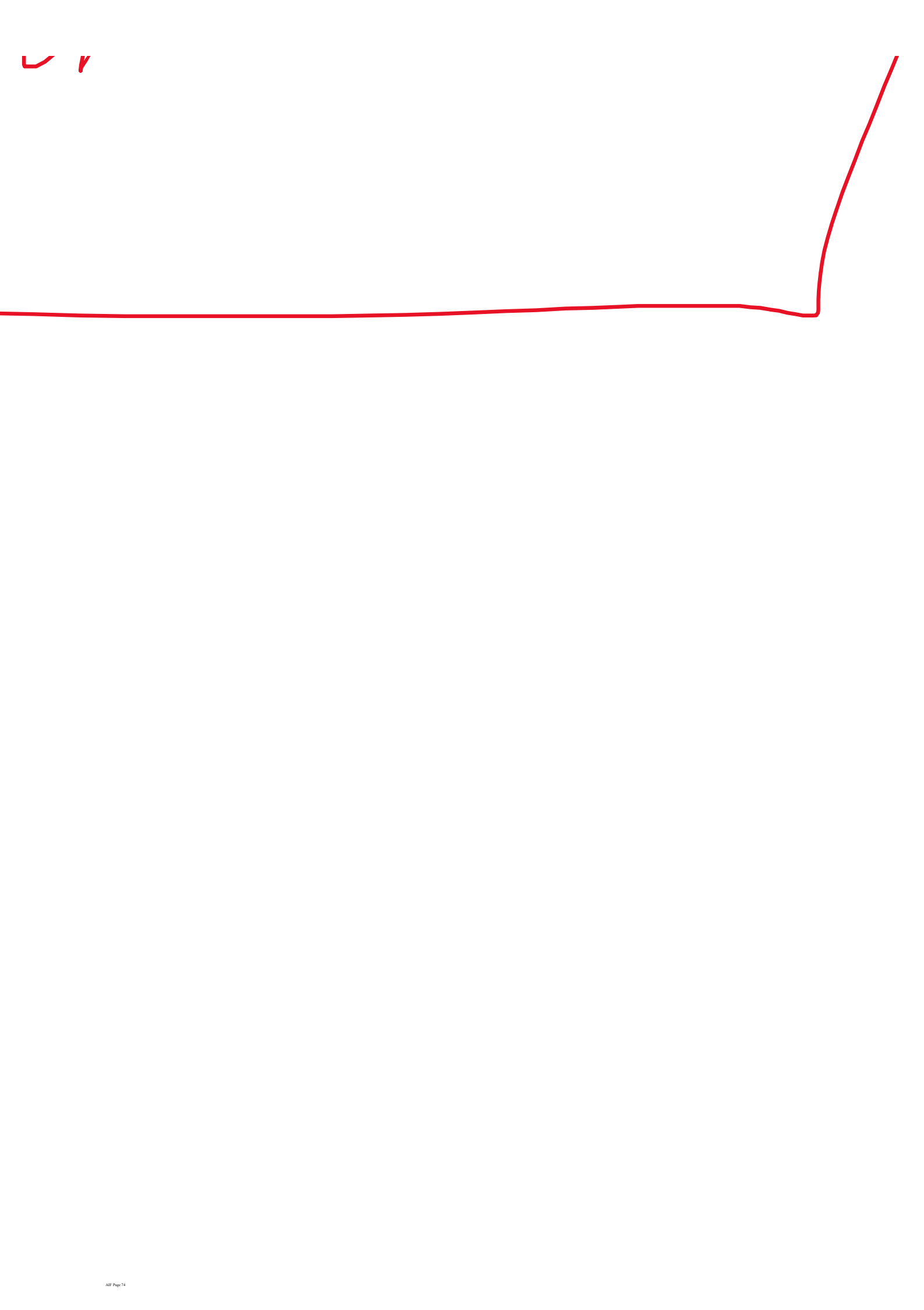


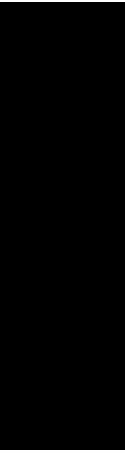
1

an

a

5-1/2 u





vector:

a) $\langle -3, 4 \rangle$

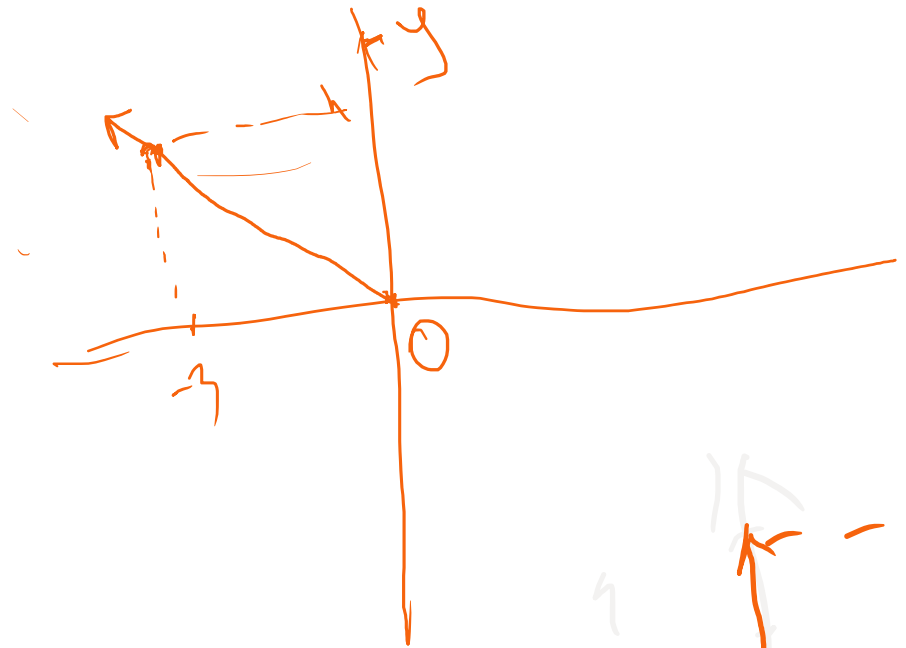
b) $\langle 2, 6 \rangle$

$\sqrt{25}$

$\|\vec{v}\| = \sqrt{a^2 + b^2}$

$\tan \theta = \frac{b}{a}$

Helps us find θ



unit vector

$(1, 0)$

$(0, 1)$





- Direction of
 - How strong
-

Step 4: Gradient

Where α is learning rate

to change w

ingly w affects the error

Gradient Descent Update

$$\underline{w_{\text{new}}} = \underline{w_{\text{old}}} - \alpha \frac{dL}{dw}$$

arning rate.

0.01

~~w_{old}~~ - w_{old}

→ 5.2

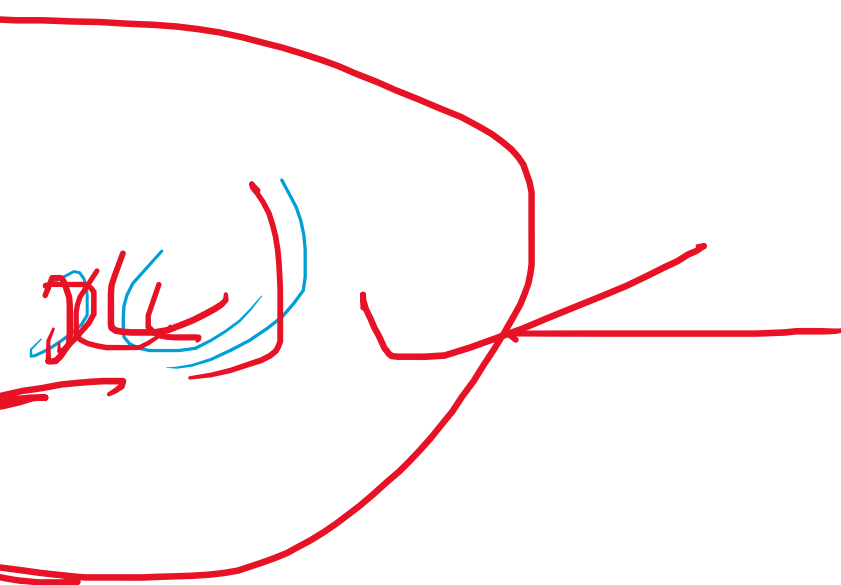
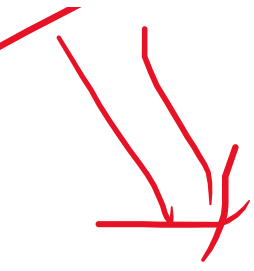
01

0.001 × 5.2

$$\frac{dw}{dw}$$

$$\rightarrow \underline{y} = \underline{(y - w)}$$

$$\underline{L} = \underline{y^2}$$



$$\frac{y_A - y}{dx^2} = \frac{dy}{dx}$$

$$= 2x$$





✓ $y = f(x) = x^2$

$f(1) = 3$

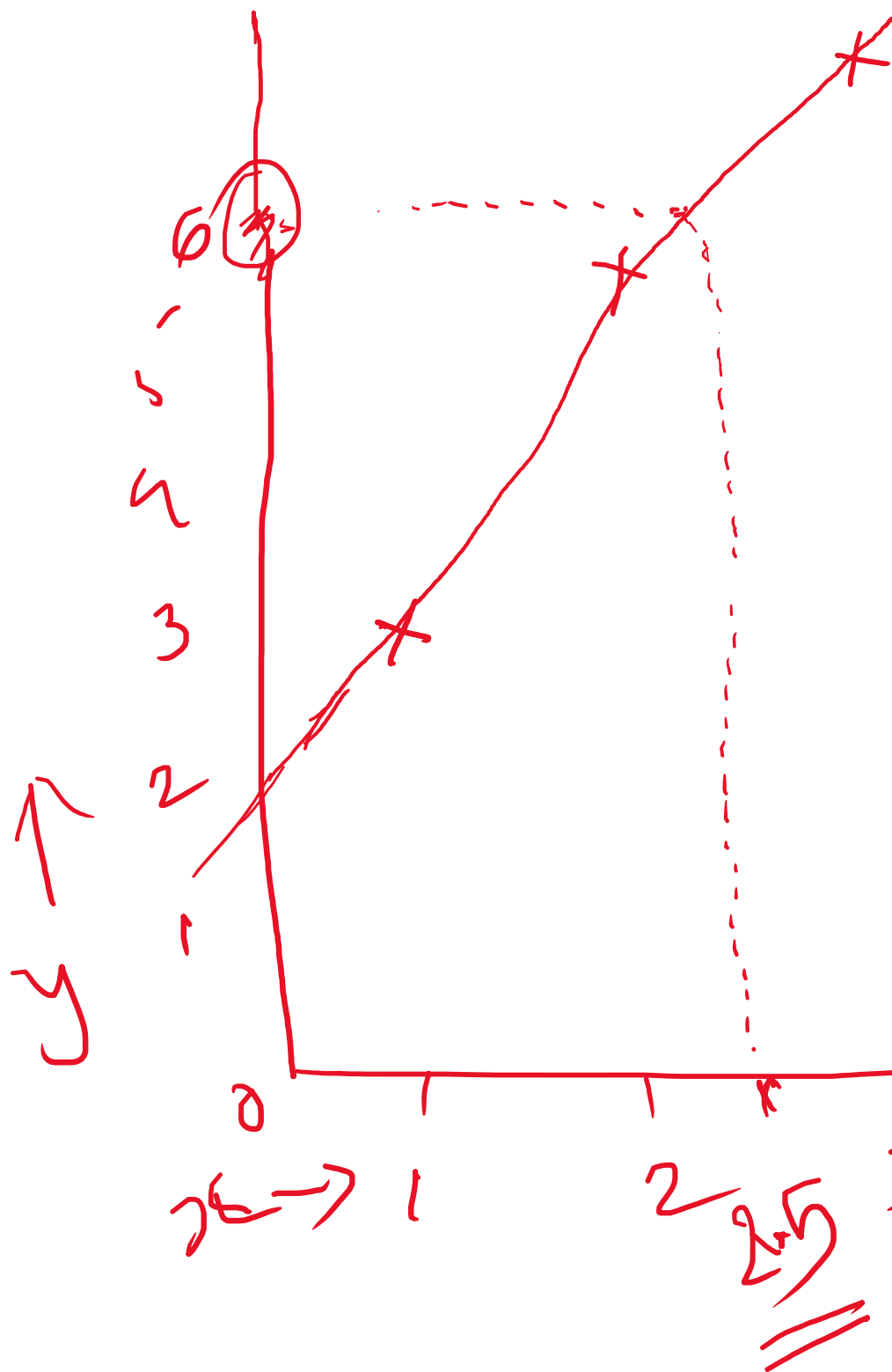
$f(2) = 6$

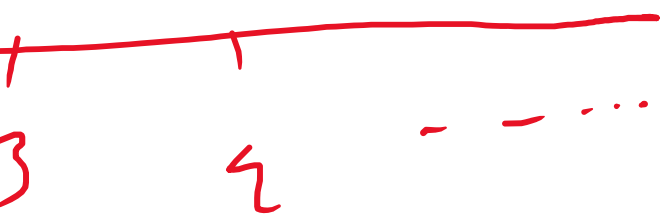
$f(3) = 11$

$f(4) = 16$

+2

$$\frac{dy}{dx}$$





$\rightarrow = 6$
 $\rightarrow = 5.$

0

9948

~~—~~

$$\frac{dL}{du}$$

—
—

2C

~~—~~

$$\frac{d\text{[scribble]}}{d\text{[scribble]}}$$

—
—

—

$$y - i\omega x)$$

$$\frac{d^4}{d\omega^4} =$$

$$\frac{d}{dx} (g - wx)$$

$$\frac{dw}{dx}$$

$$\frac{d}{dx} (g - wx)$$

2021

$$\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{dx} \cdot \frac{dx}{dx}$$

$$y = \sin x$$

$\frac{dy}{dx}$ = rate of change
of y w.r.t x

$\Delta x \rightarrow \text{small}$

$0.999 \rightarrow 1.000$

rate of change of
of y w.r.t x , x changes = $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$
infinitely small

$f(x) \Rightarrow 1$

$y = x \Rightarrow 1$

of

limit
 $\Delta x \rightarrow 0$

$$\frac{\Delta y}{\Delta x}$$

dw

$$\frac{dL}{dw}$$

=

$$\frac{dL}{d\mu}$$

=

$$2C_y$$

$$\frac{dy}{dx}$$

$$(-\omega x) + (-x)$$

$$\frac{1}{4} \omega x$$

dy

dw

O

—

—

du

dc

x

=

W
Z

→

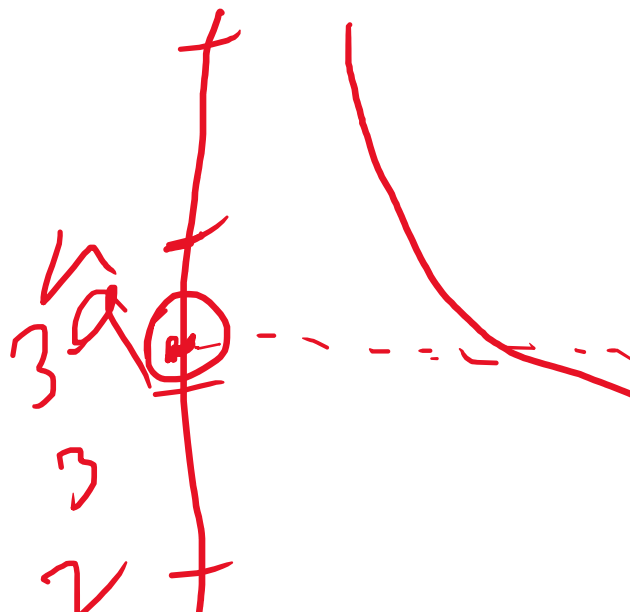
0

— JL

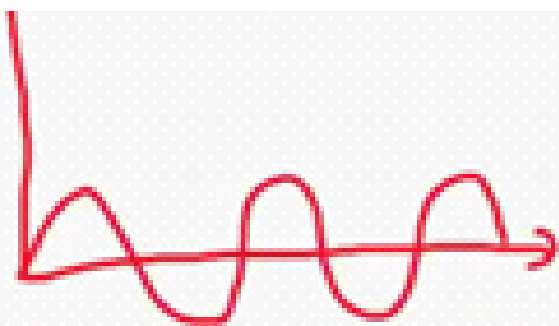
—

$$\underline{\underline{y = \sin x}}$$

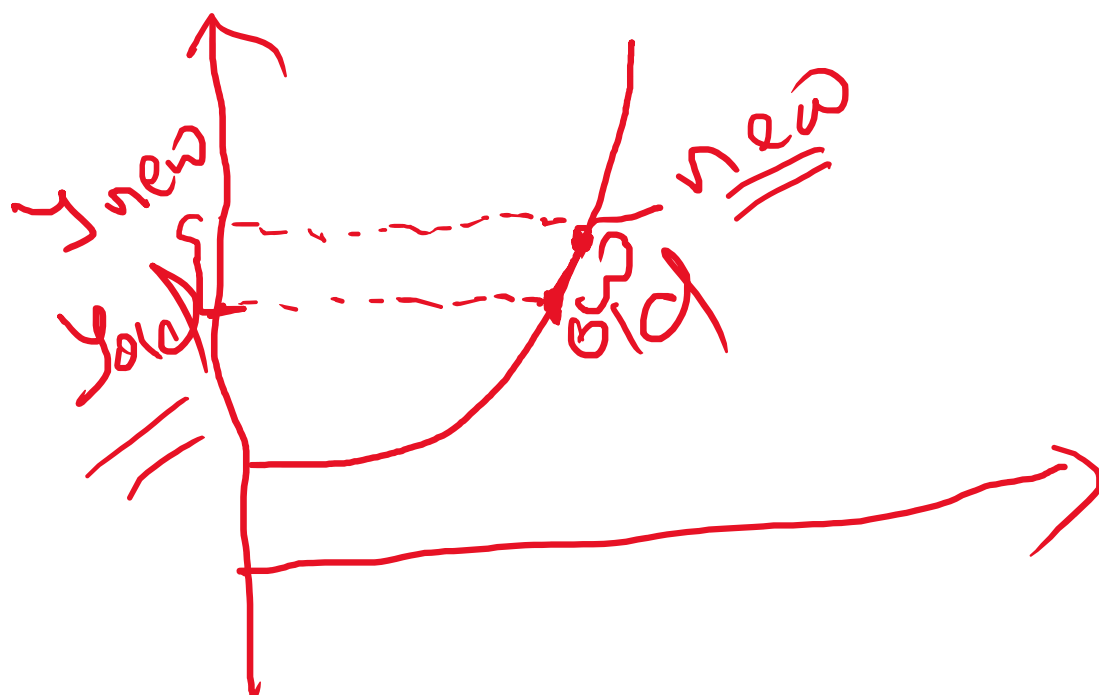
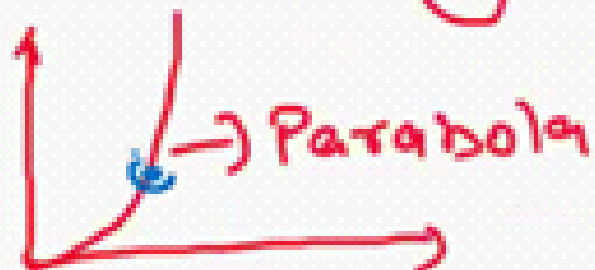
$$\underline{\underline{y = x^2}}$$



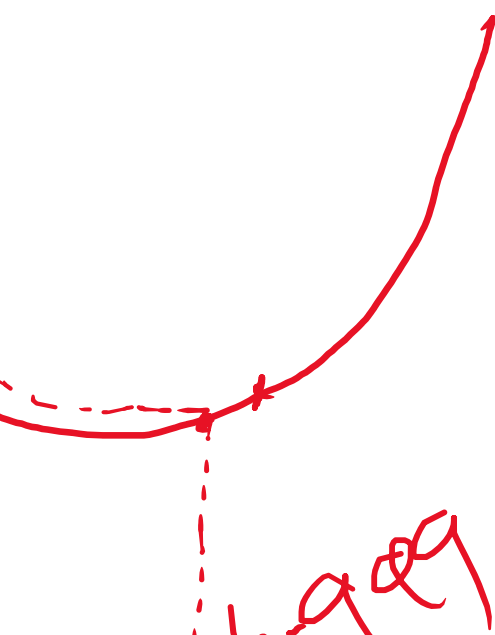
$(\pi) =)$



d



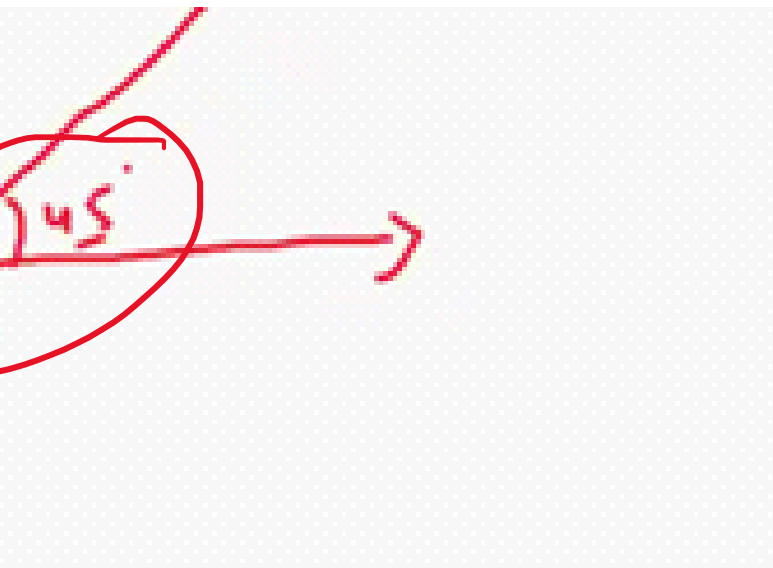
x_0



$$\frac{dy}{dx}$$

$x =$

$\Delta x =$



old - x_{new}

4.999 ✓

x_{old} - x_{new}
add

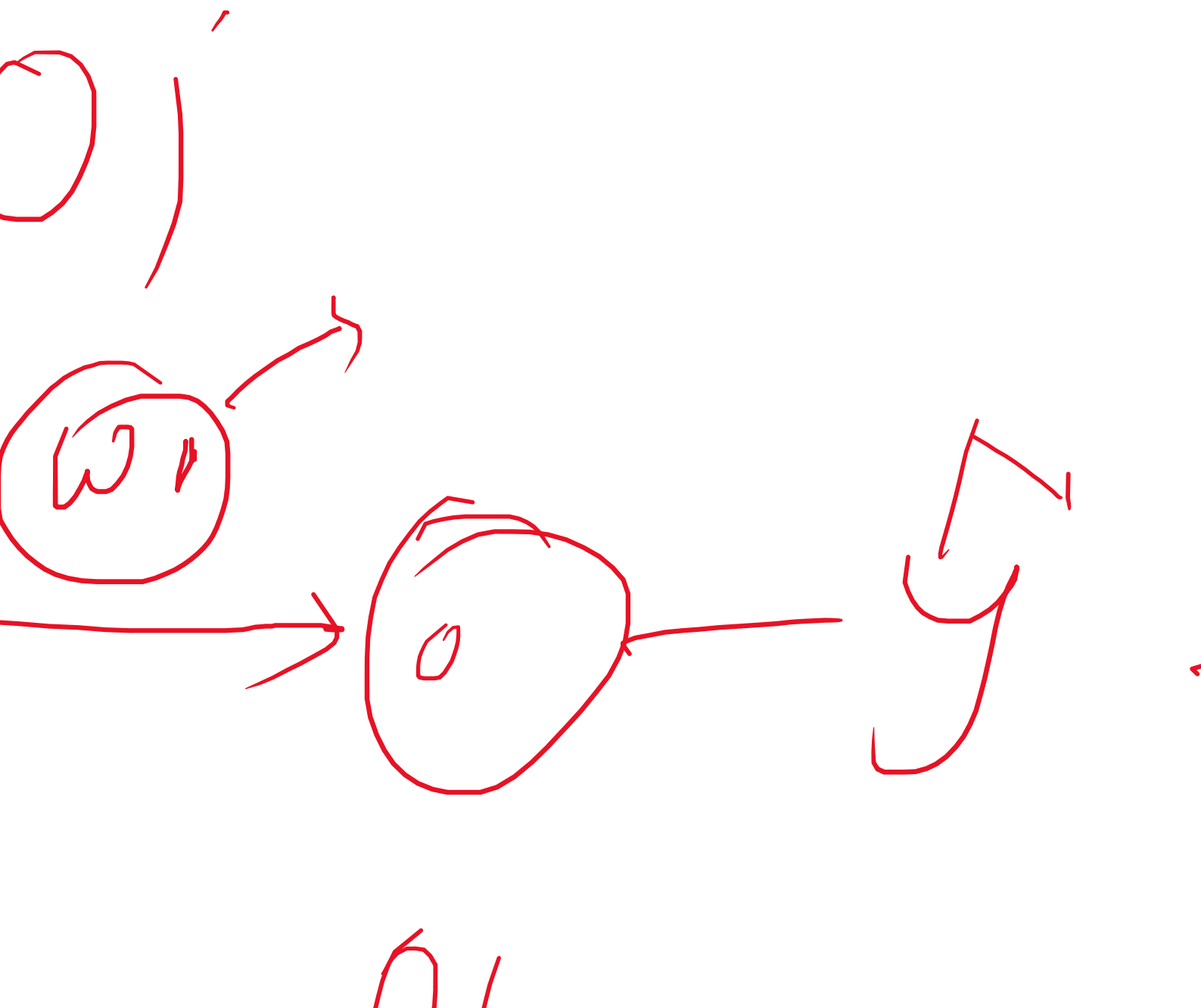
$$\frac{\Delta y}{\Delta x} = \frac{0.1}{0.001} = 100$$

220

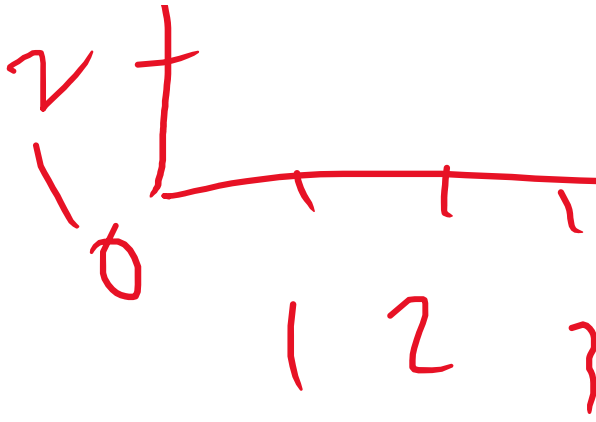
A collection of red hand-drawn scribbles on a white background. The scribbles include a large circle with an 'X' inside, a large 'X' itself, and various loops, lines, and abstract shapes. Some of the shapes resemble stylized letters or symbols, but they are not clearly defined. The overall impression is of a series of quick, gestural marks made with a red pen or marker.

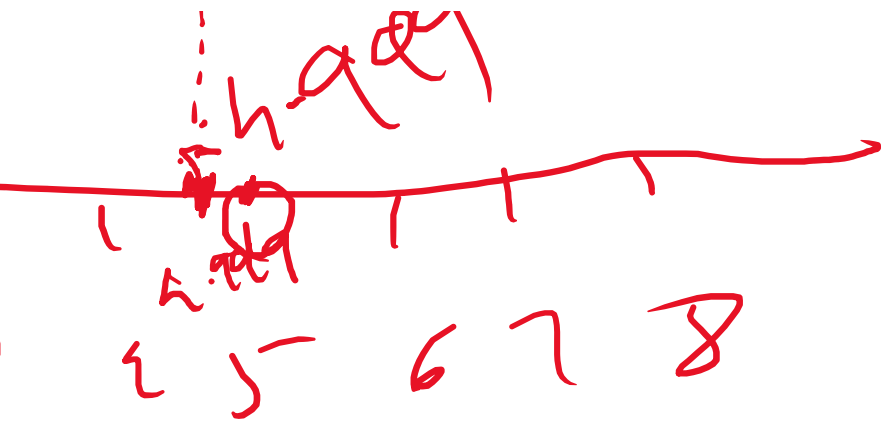
WY - WU





$$\frac{dL}{d\omega}$$





$$\Delta x =$$

$$\Delta x =$$

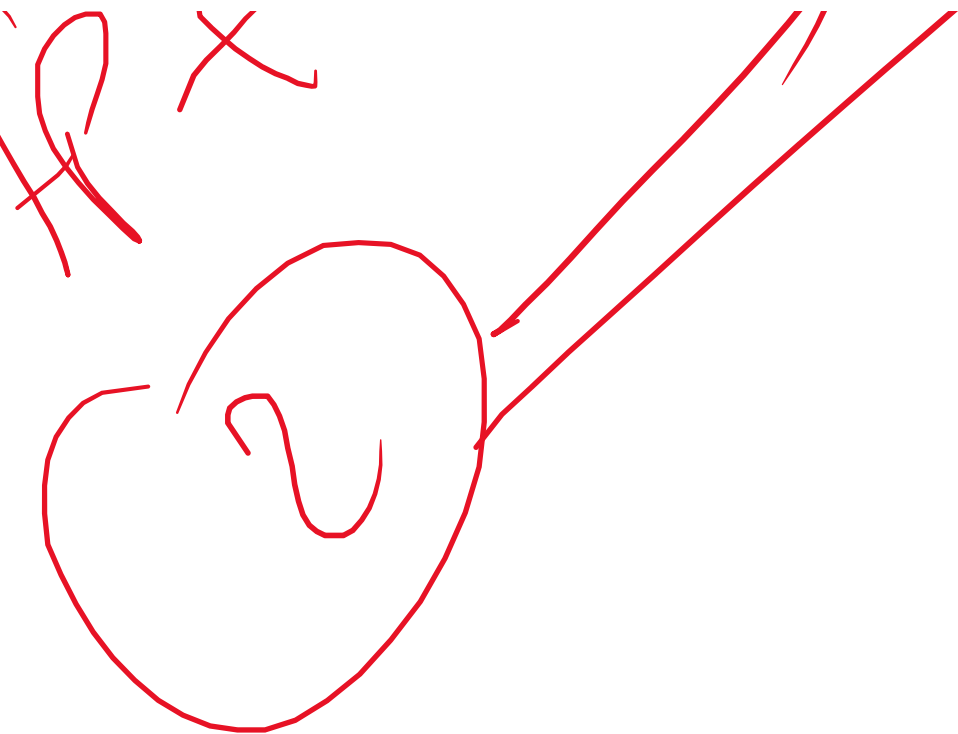
$$\Delta x = \text{old} - \text{new}$$

$$\begin{array}{r} 0.014 \\ \hline 5 - \bar{h}.999 \end{array}$$

$$\begin{array}{r} 0.001 \\ \hline \hline \end{array}$$

$$0 = 4 - 3.9 = \cancel{0.1} 0.1$$







dL

da

$\frac{dL}{do_1}$

do_1

0/p
doi
dwp

$$f(x) \quad y =$$

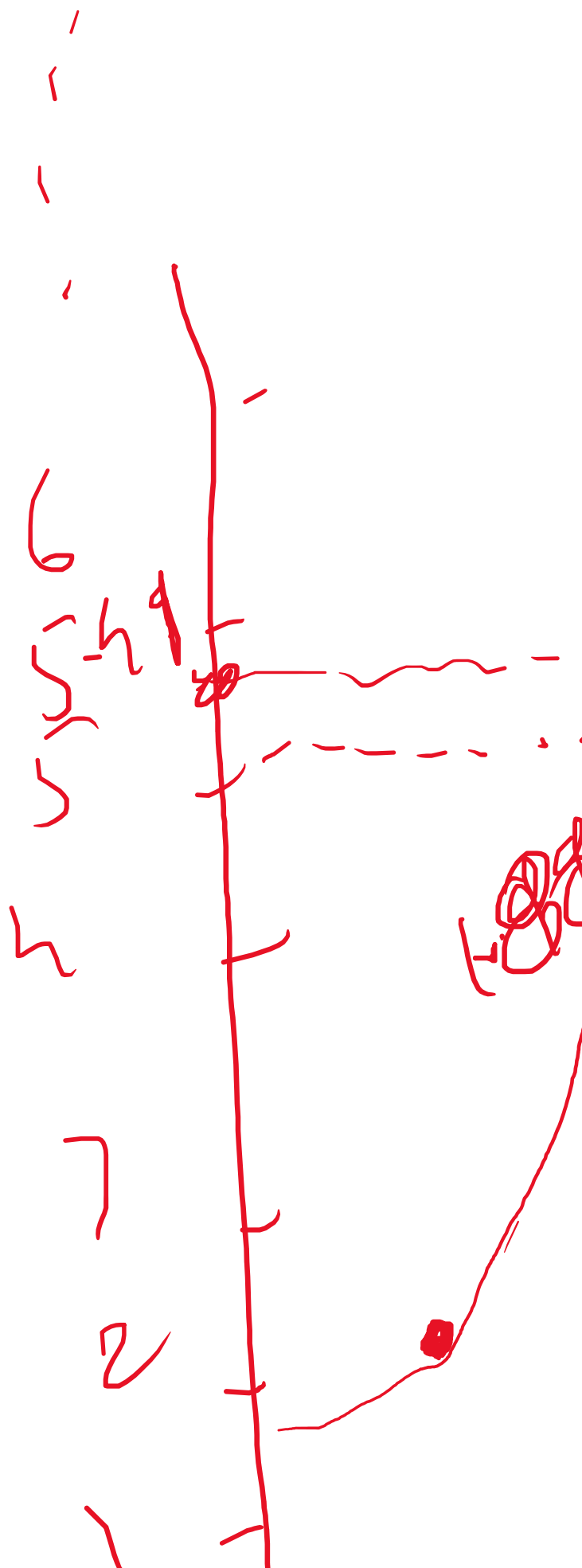
$$y = f(1) =$$

$$y = f(2)$$

$$\underline{x^2 + 1}$$

2

5





$$X_{old} =$$

$$X_{new} =$$

$$Y_{old} =$$

$$Y_{new} =$$

2

2.1

5

5.41

$$\Delta x = \underline{\underline{-0.1}}$$

$$\Delta y = -0.4$$

$$\frac{\Delta y}{\Delta x} = \frac{-0.4}{-0.1}$$

$$= \underline{\underline{4}}$$

-1

7 ~~8~~

21
—

0.1

$$y = f(3)$$

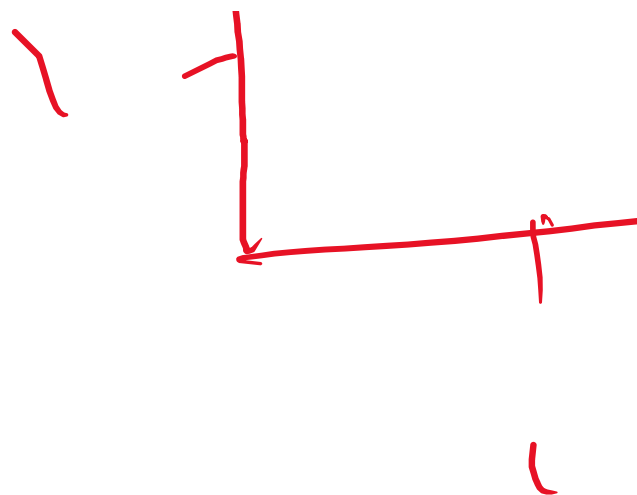
$$y = f(2.1)$$

$$f(0.8)$$

$$= 10$$

$$) = 5.41$$

8)



X new
 X new
~~new~~
~~new~~

$$d=2$$

$$d=5$$

$$A_x = 0.12$$

$$\Delta y = 0.4656$$

dy

$$y = \dots$$

$$y = f(x)$$

$$f(x)$$

$$f(x)$$

x

$\checkmark$

$$= x$$

$$1) = y = 1$$

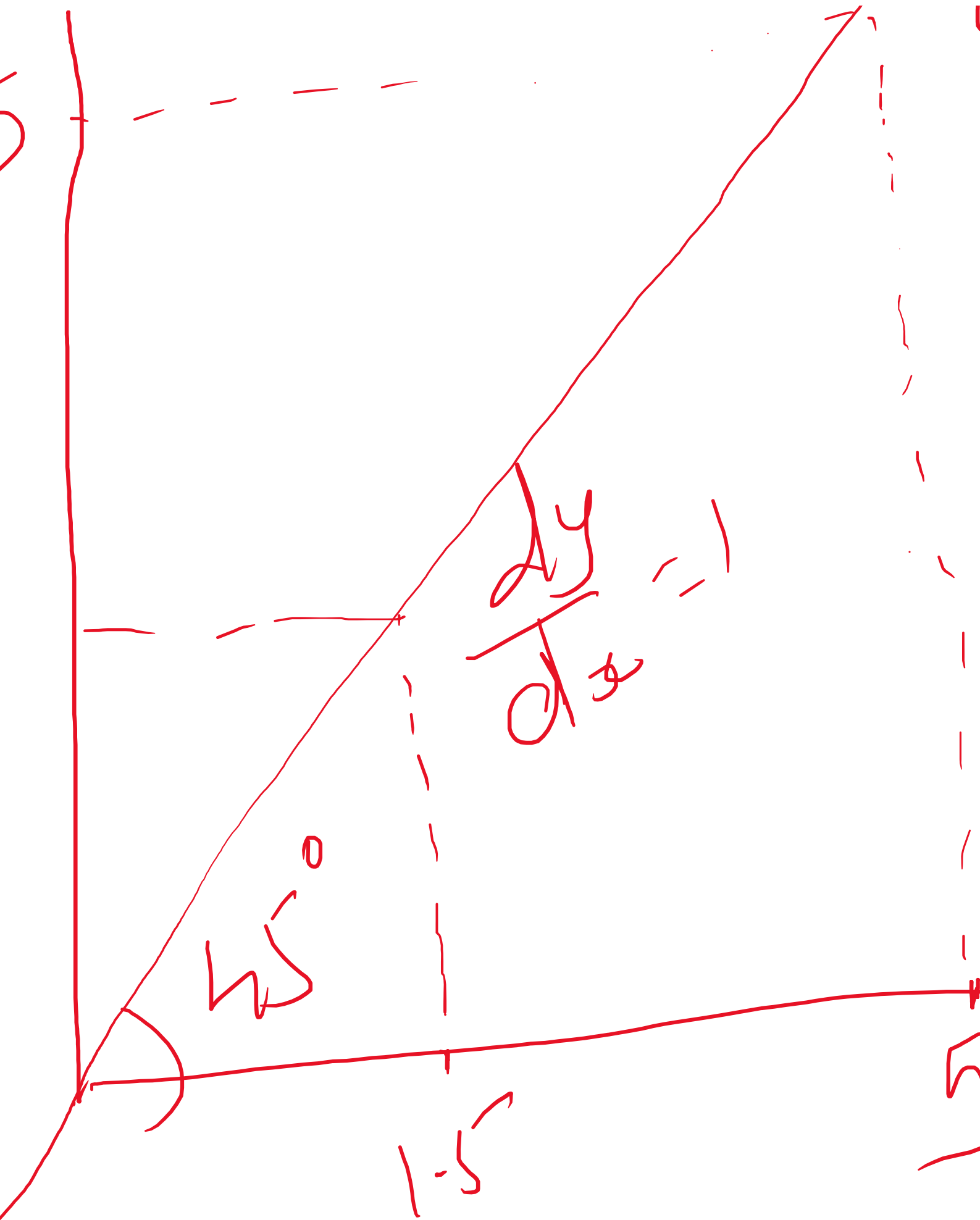
$$2) = y = 2$$

$n \setminus$

4

2

$\checkmark$

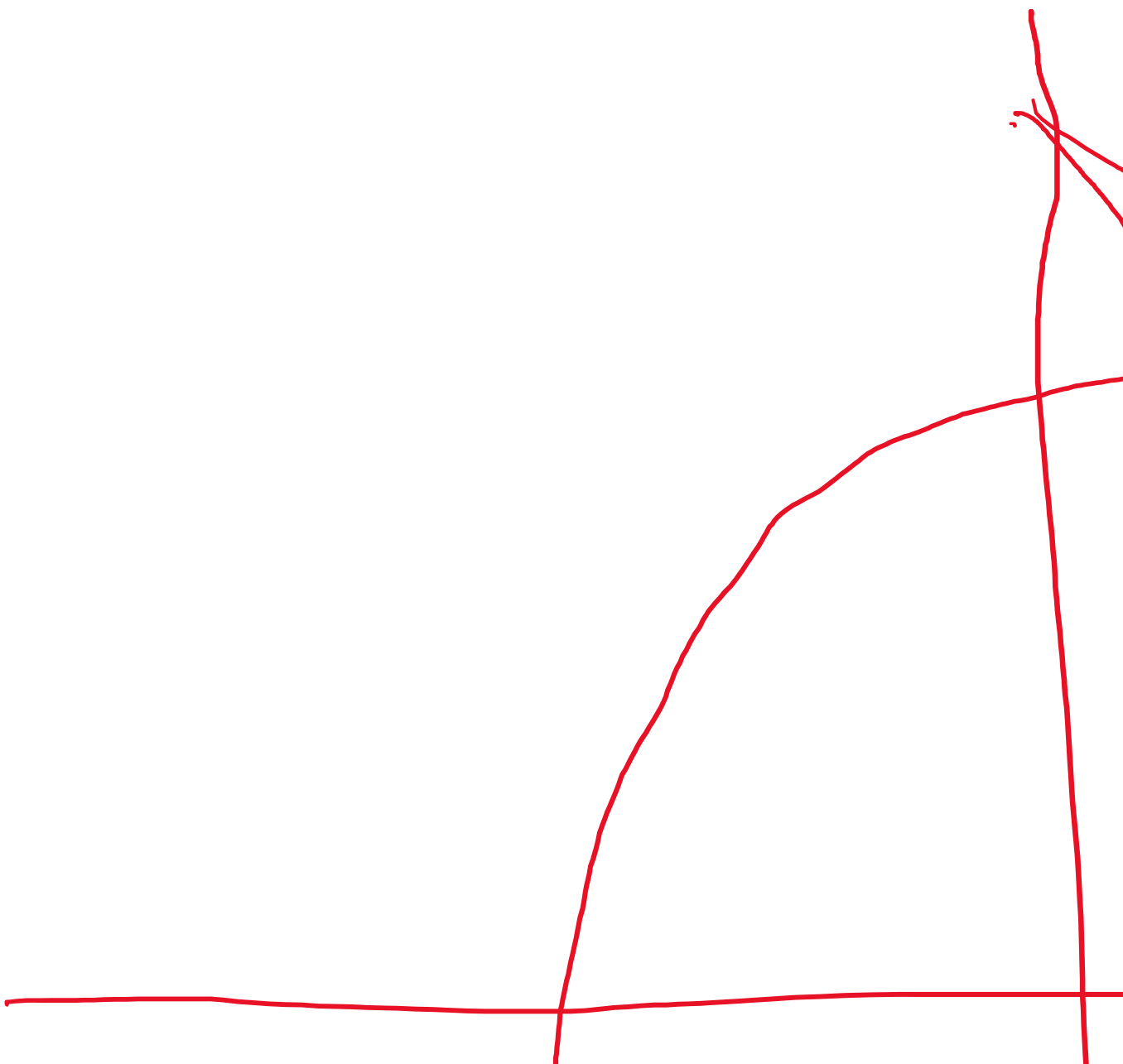


uv
ds

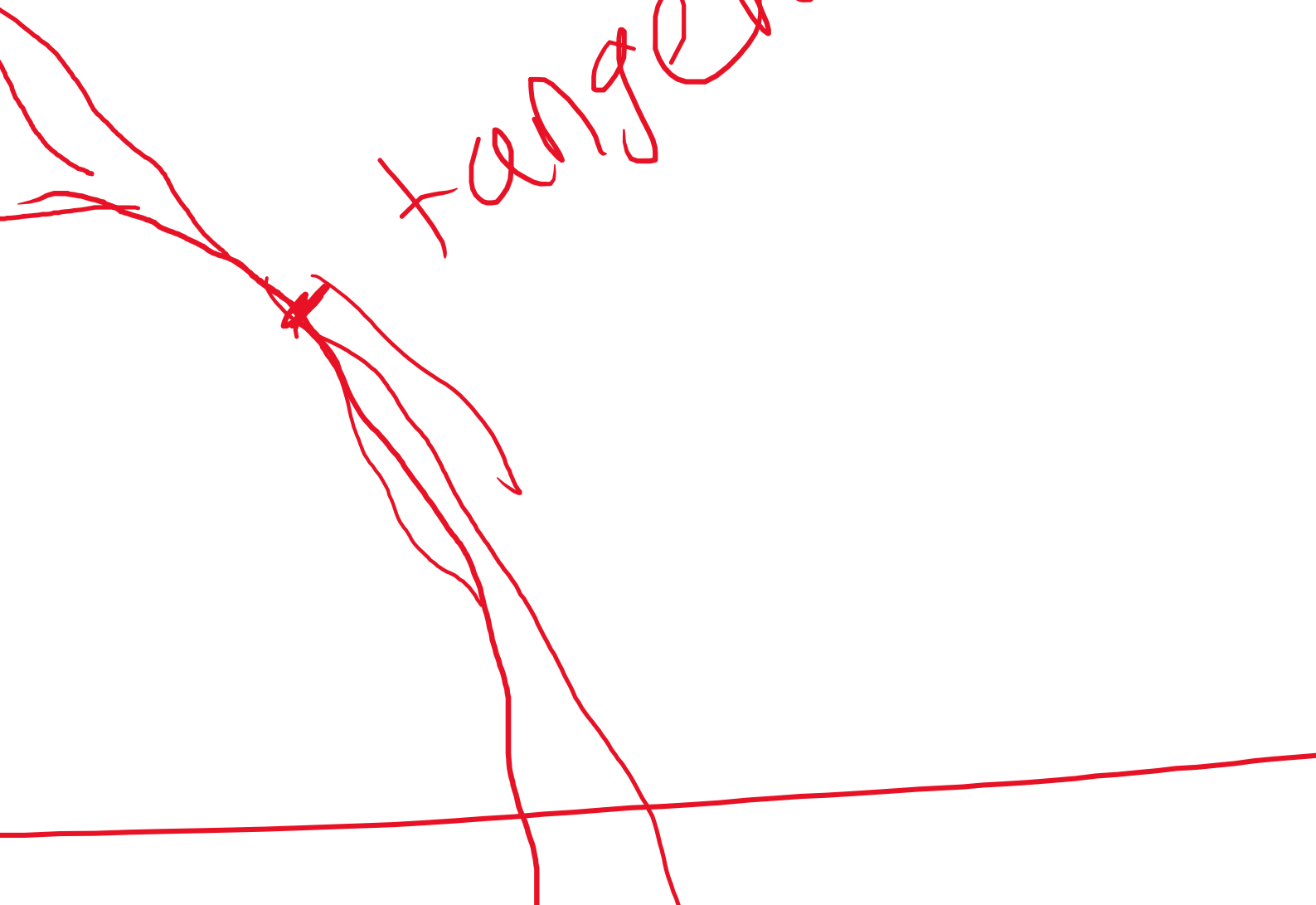
.

FC

$$3) \Rightarrow y = 3$$



tangent

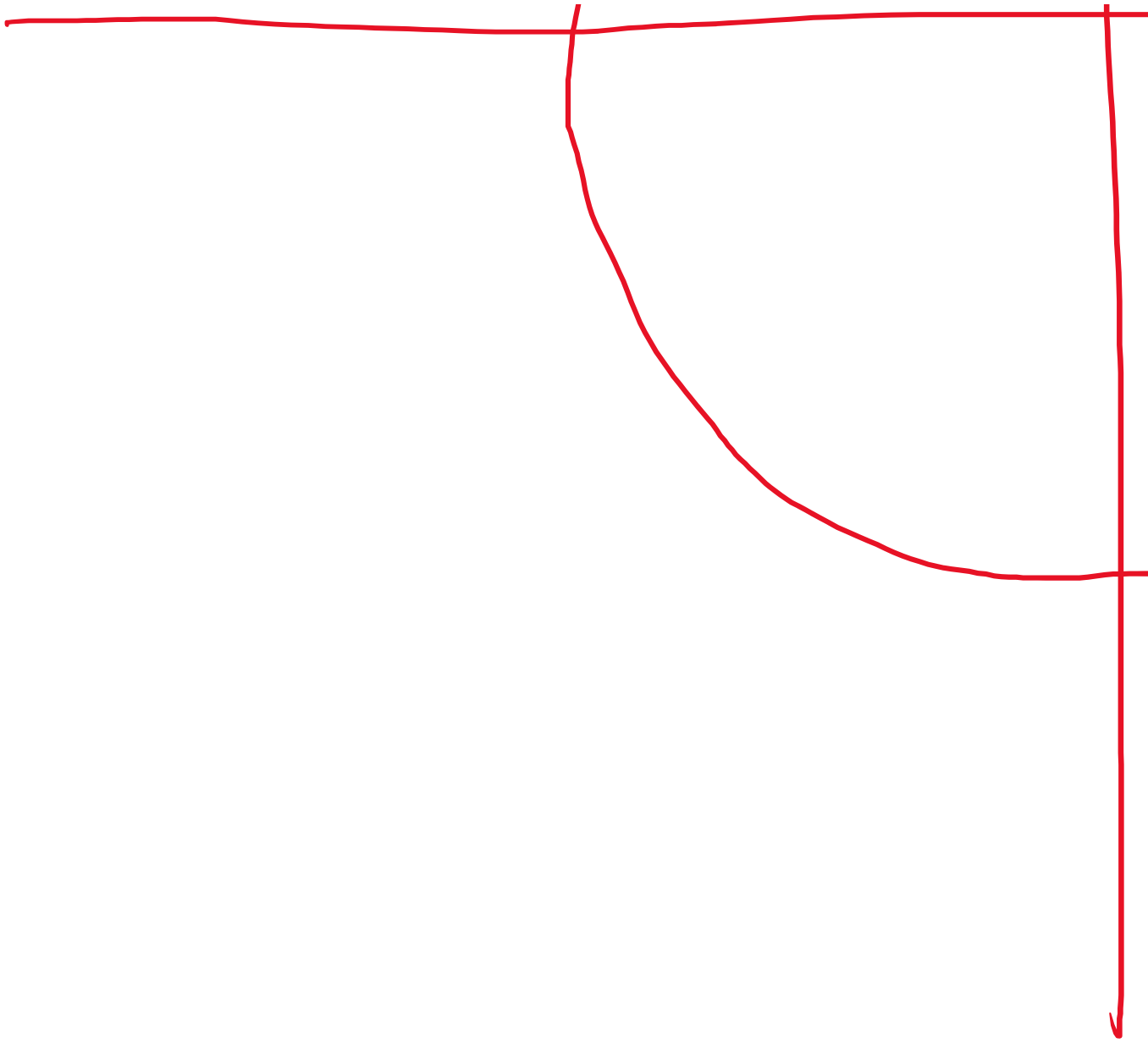


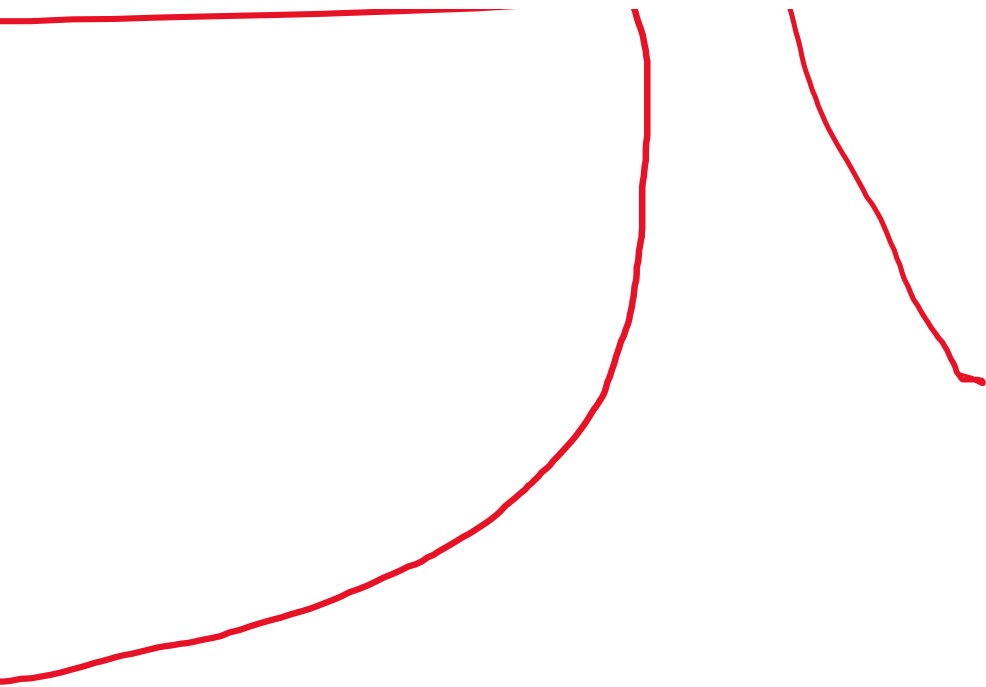
→

SV =

$$\frac{x^2 + 2}{\sin(x)} \quad \text{un}$$

ivate
fan²





U

2

1

1

1

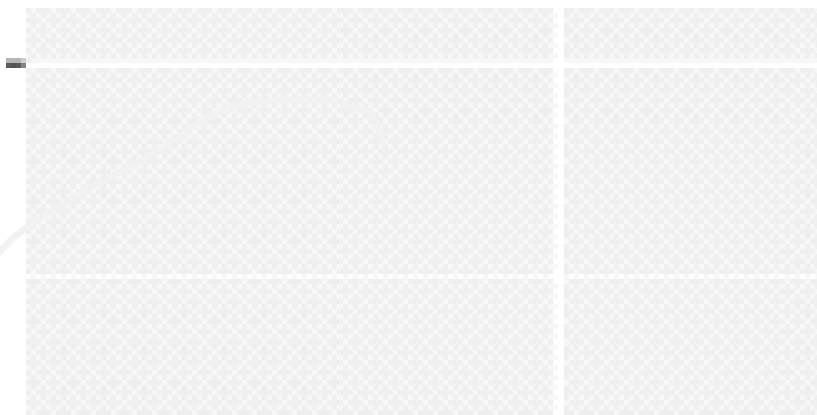
3

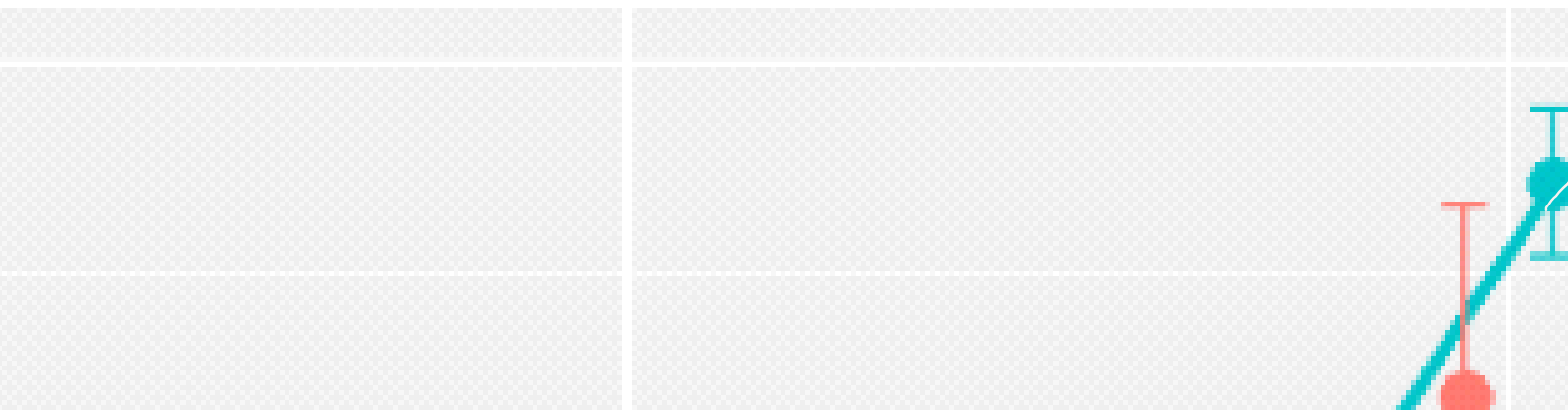
1

$$2x + 3$$

$$2x + 2x + 5$$

6









$$0 = \frac{dy}{dx}$$



10



31

3D

$$y = 2x,$$

$$\underline{2x} = 1$$

$$\underline{\underline{2x}} = 1$$

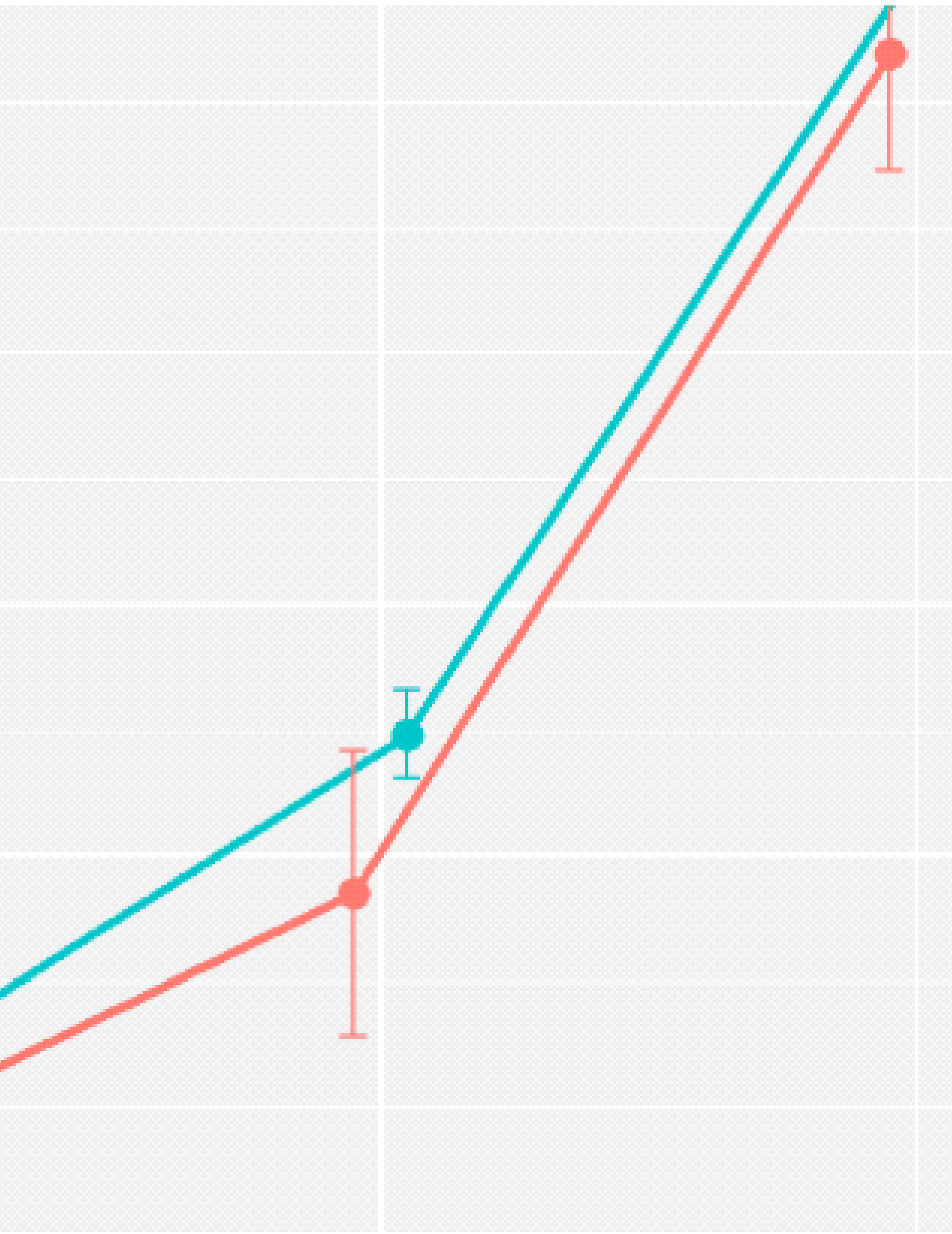
$$+ 3x^2 + 5$$

$$x_1 = 2$$

$$x_2 = -2$$

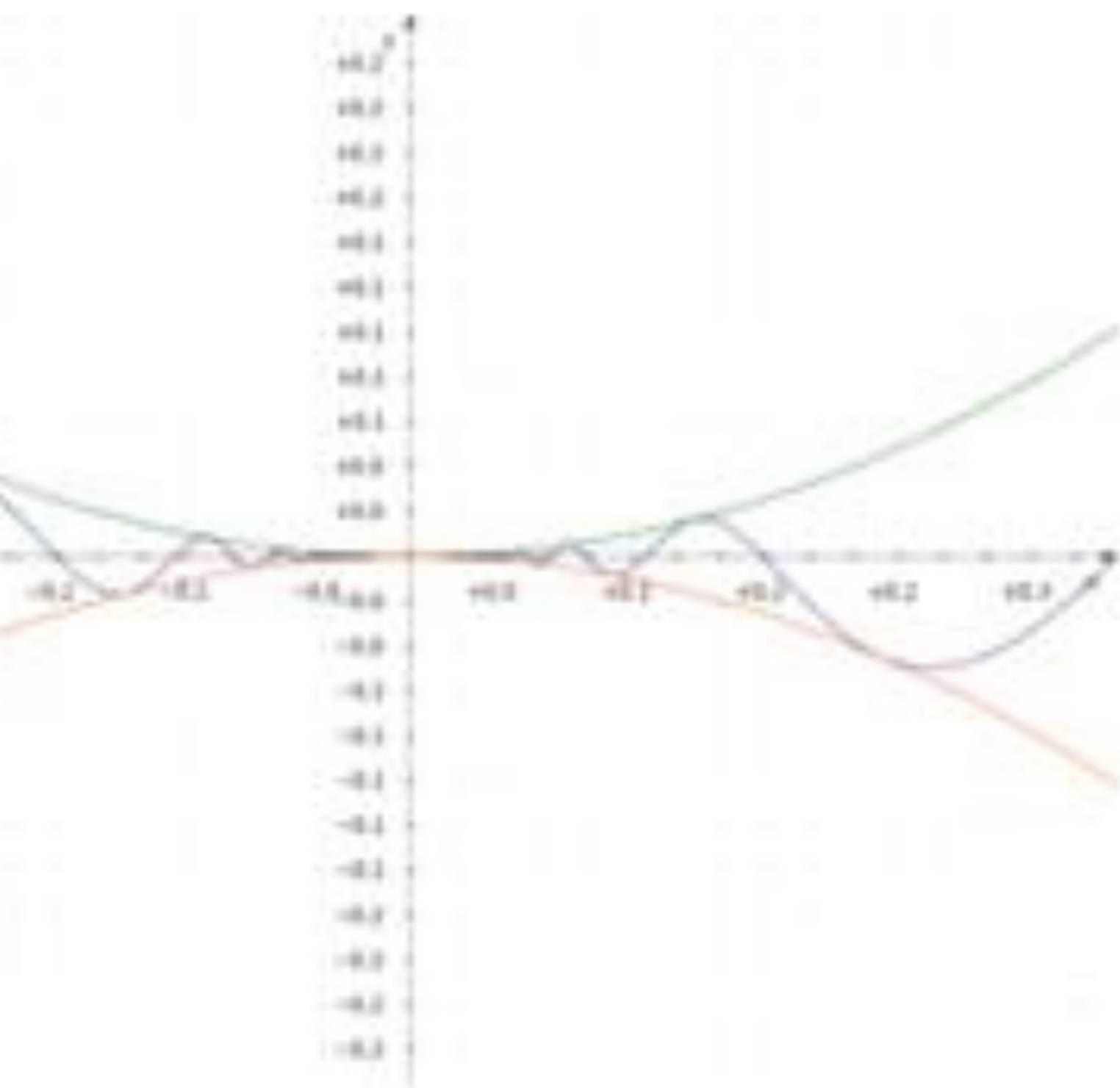
$$4 - 15$$





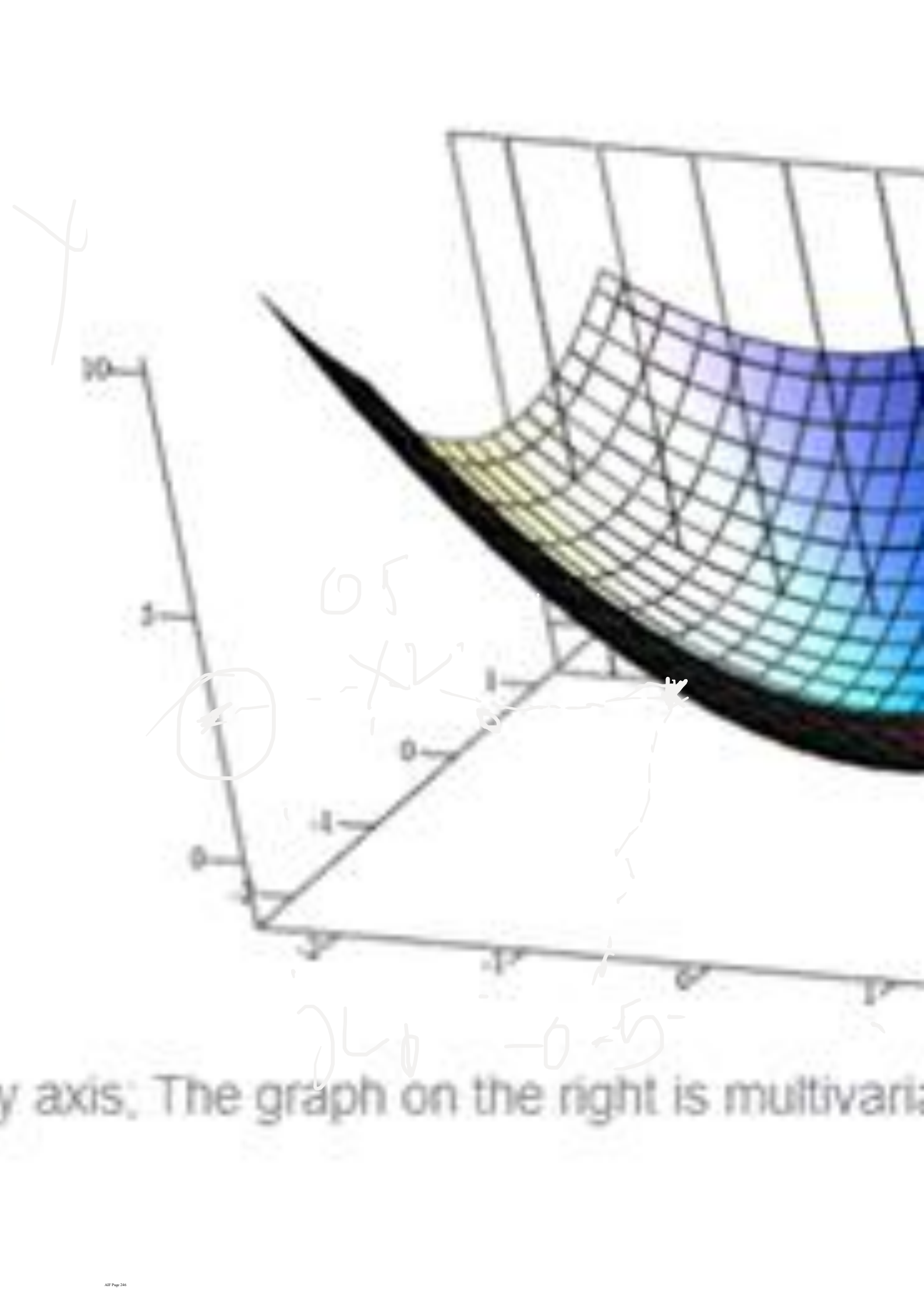


Functions of
a x, y, z axis

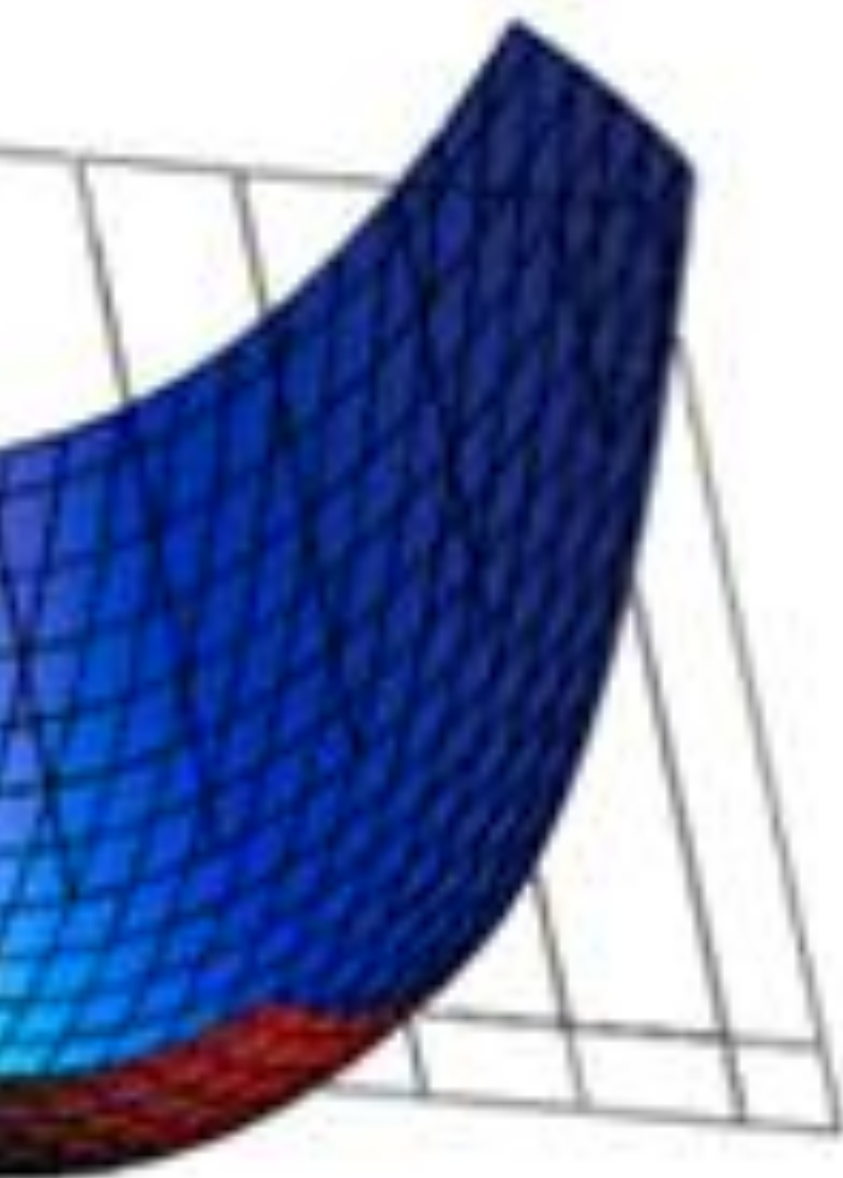


of one variable (left) are graphed on an x-y

5.



y axis. The graph on the right is multivariable



ate and is graphed on

$$\hat{y} = 10$$

$$y = 15$$

XG
/

24

3 ✓

3

$$y = x^2 + 2$$

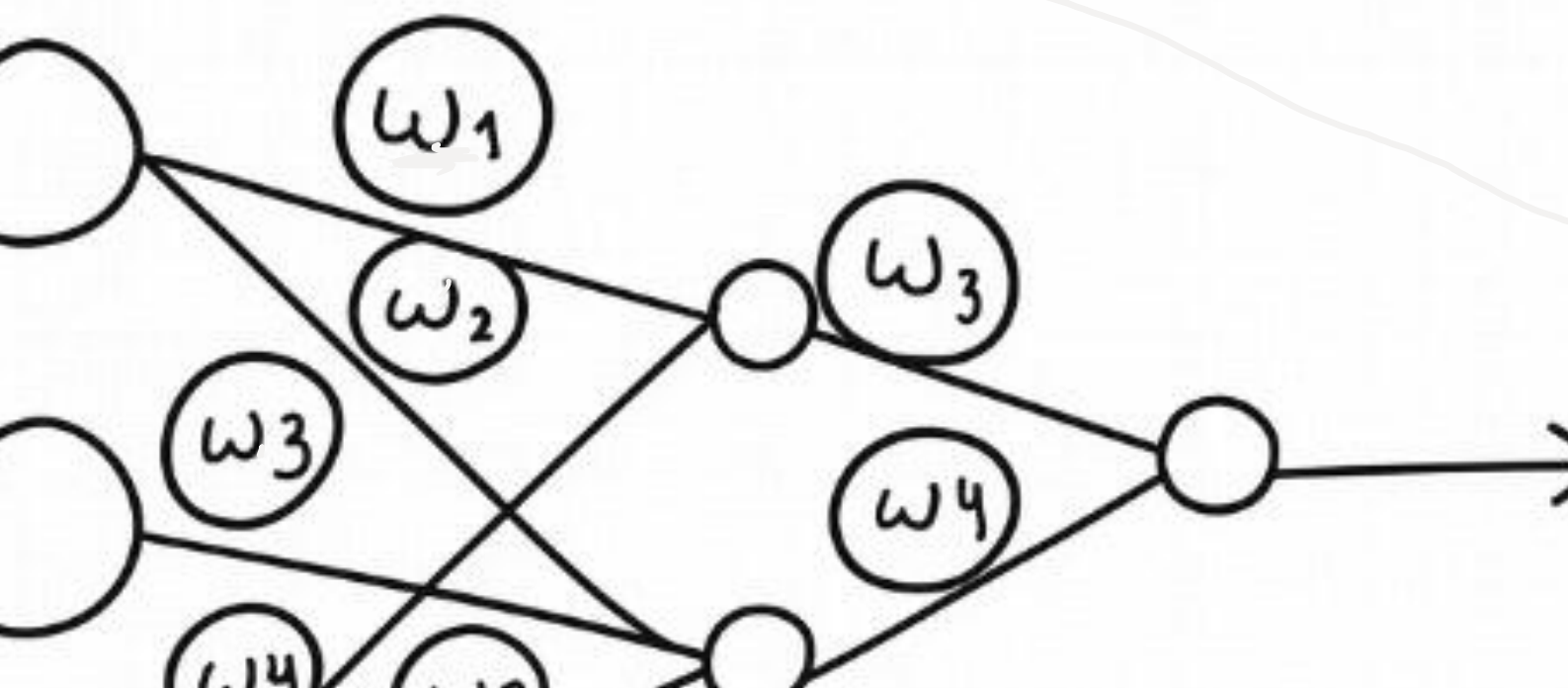
$$y = \sin x$$

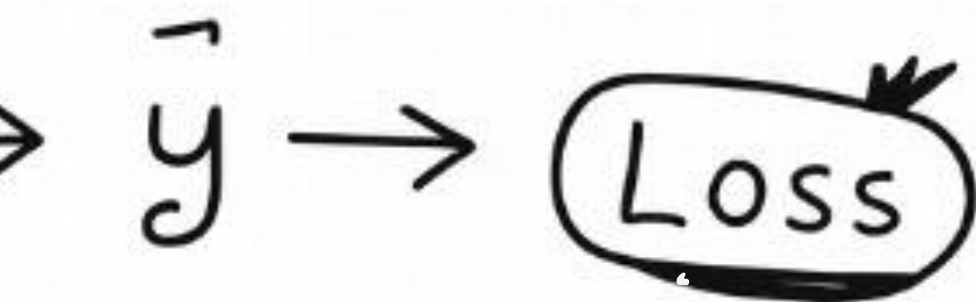
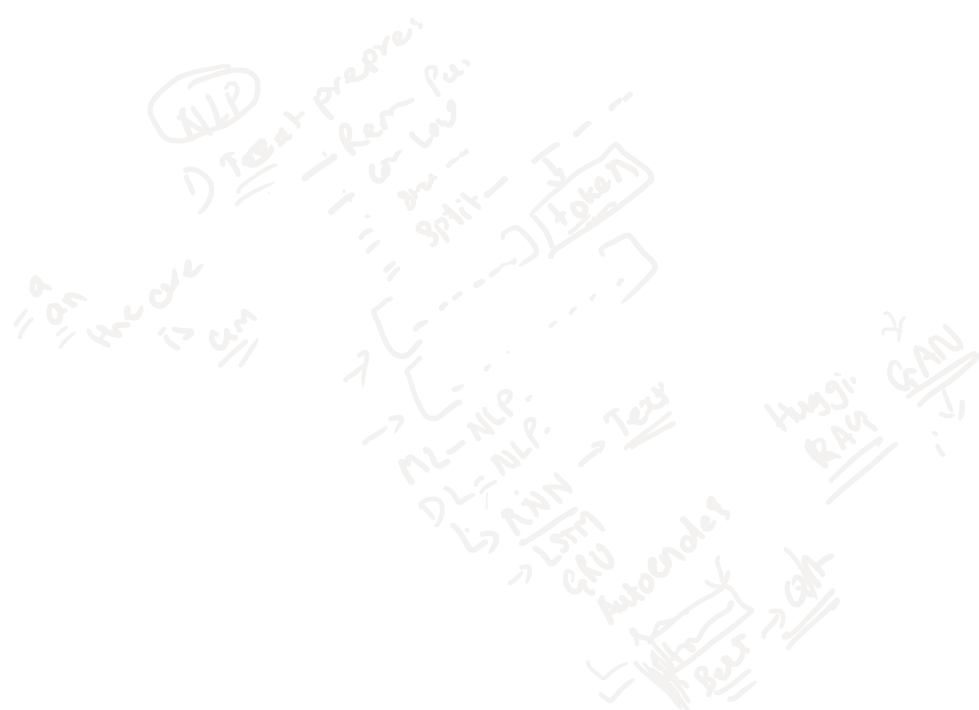
$$y = 3x + 2$$

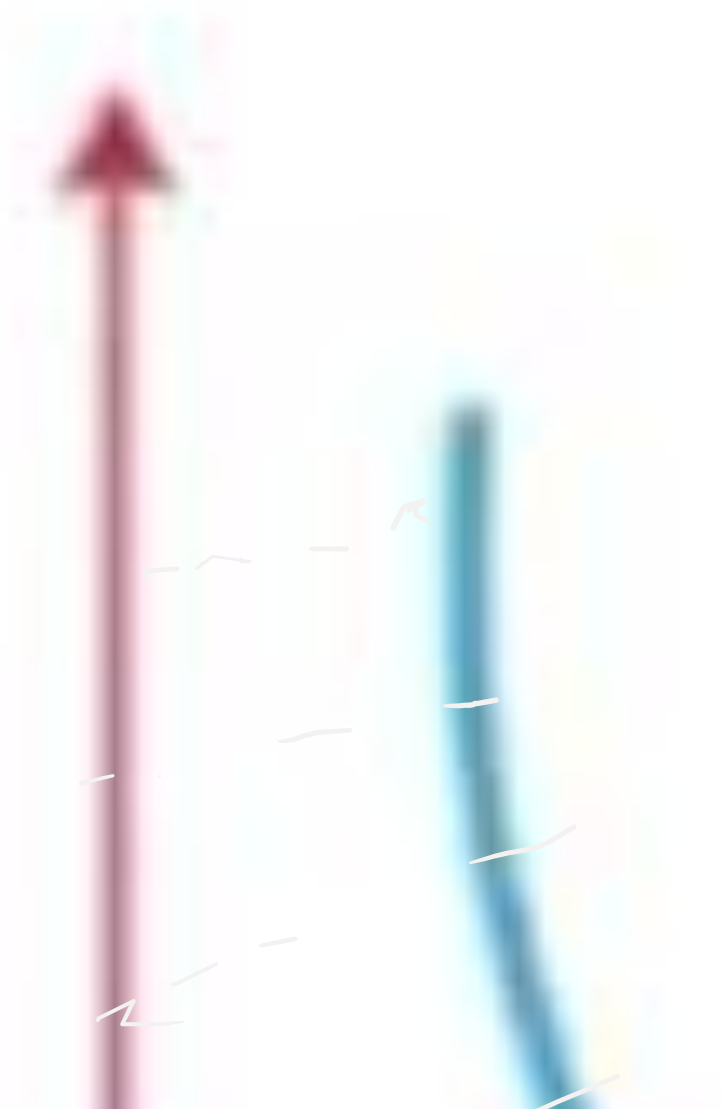
Univariable

$$y = 2x_1 + 4x_2 + 3$$

Backpropagation









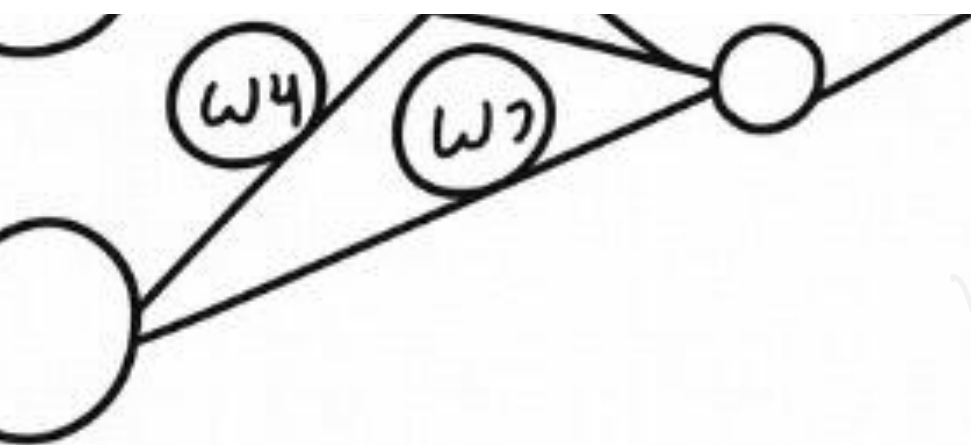
grade


$$y = 2x_1 + 4x_2 + 3$$

Bivariate

$$y = 2x_1 + 2x_2 + 3 + 5$$

Multivariate



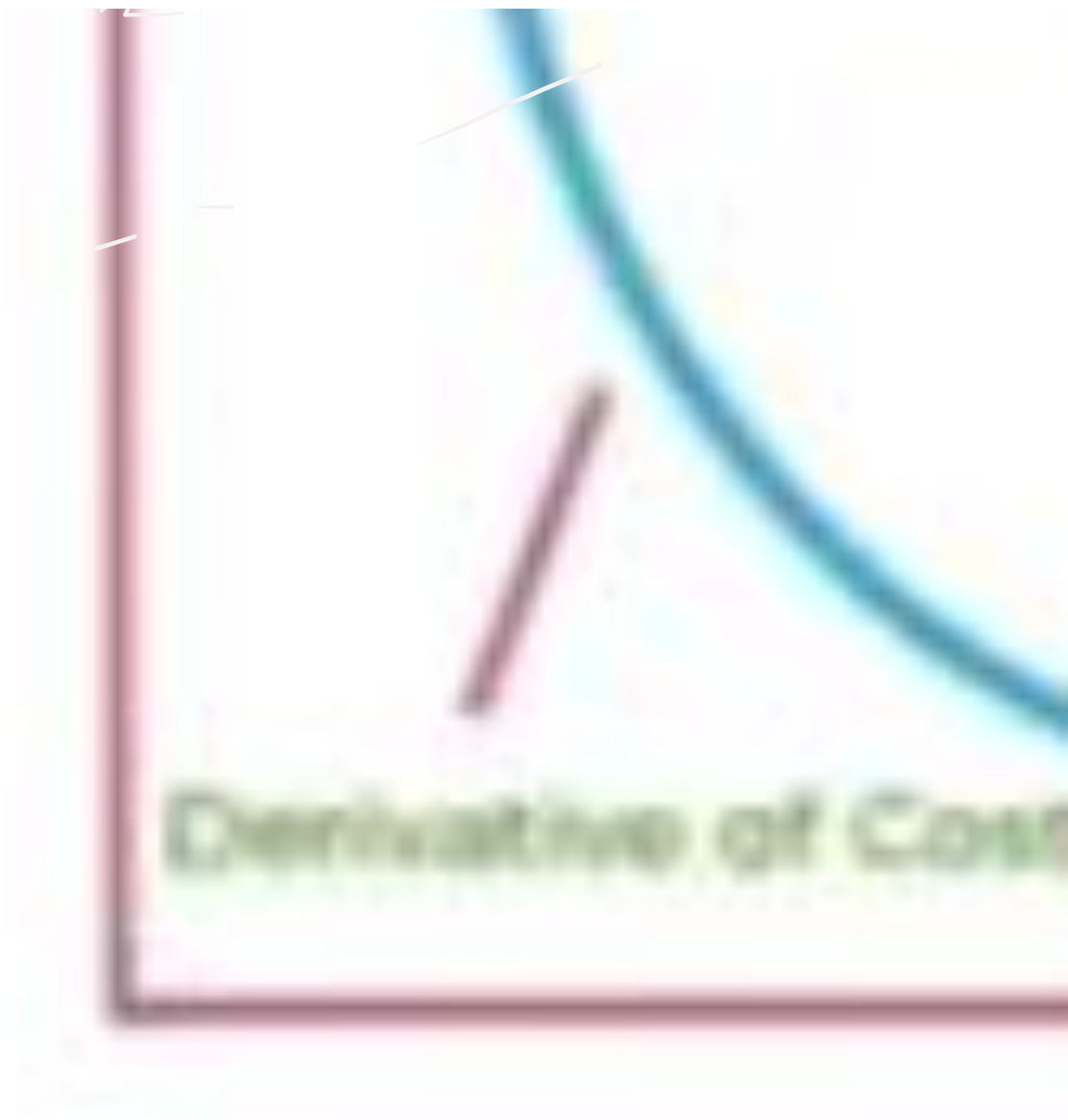
$$\frac{y_1 + x_{n_2+2} + 5}{5}$$

e

Loss function

ion ($\omega_1, \omega_2, \omega_3, \omega_4, \omega_5,$
 ω_7, ω_8)

loss $f(\omega_i)$





Weight

✓

Minimum Cost



n

n

o

14

15

1051

used minimally

2

1

→

Univariate $\rightarrow ex \rightarrow y$

Bivariate $\rightarrow ex \ y = 3n$

Multivariate $\rightarrow ex \ y \ y =$

$$y = 2x + 2 \rightarrow \text{Line or}$$

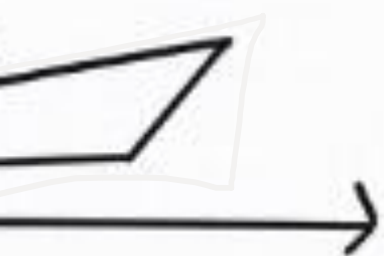


$$n_1 + 2n_2 + 2 \rightarrow \text{Plane}$$



$$2n_1 + 3n_2 + 4n_3 + 3 \rightarrow \text{Hyperf}$$

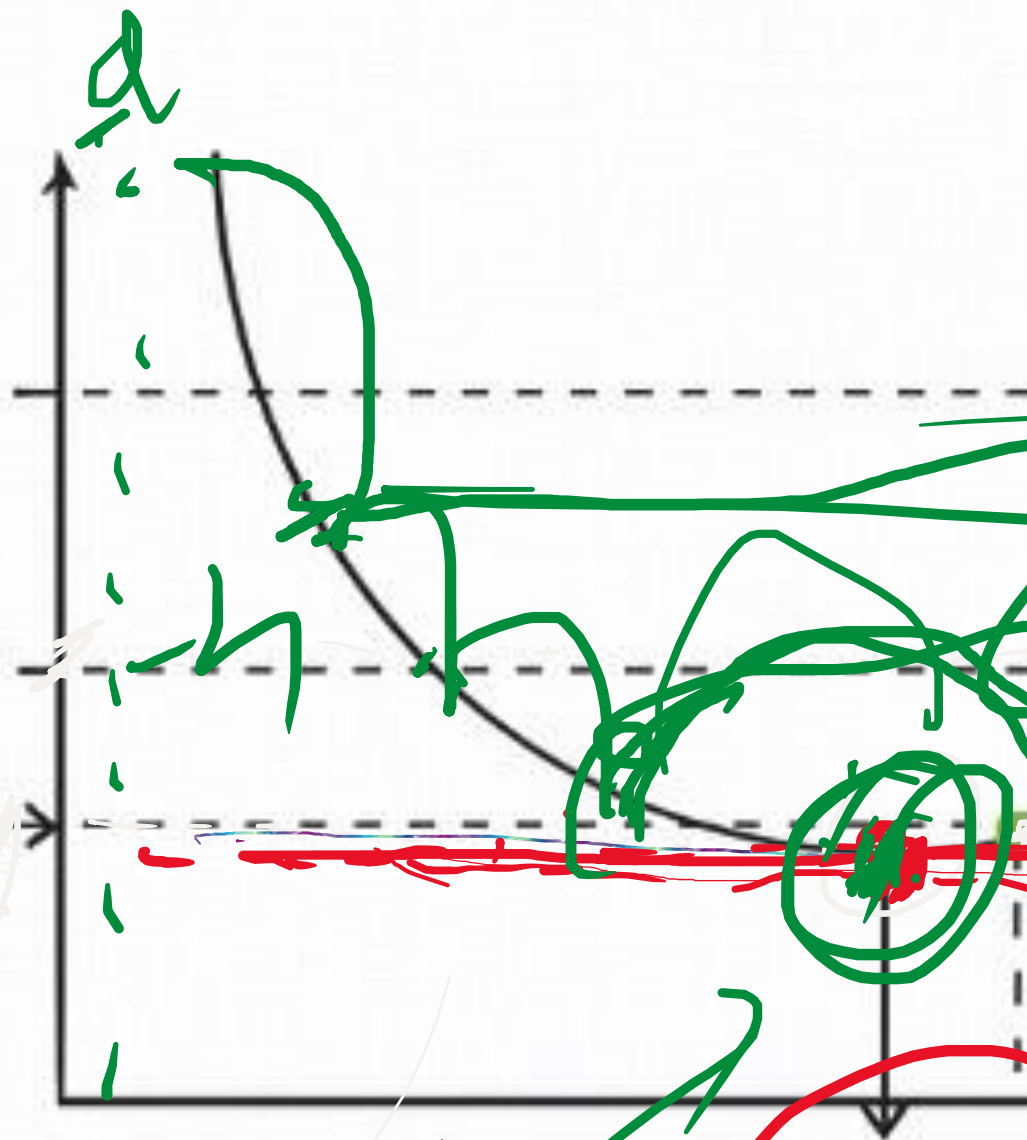
Squeezy line



Plane

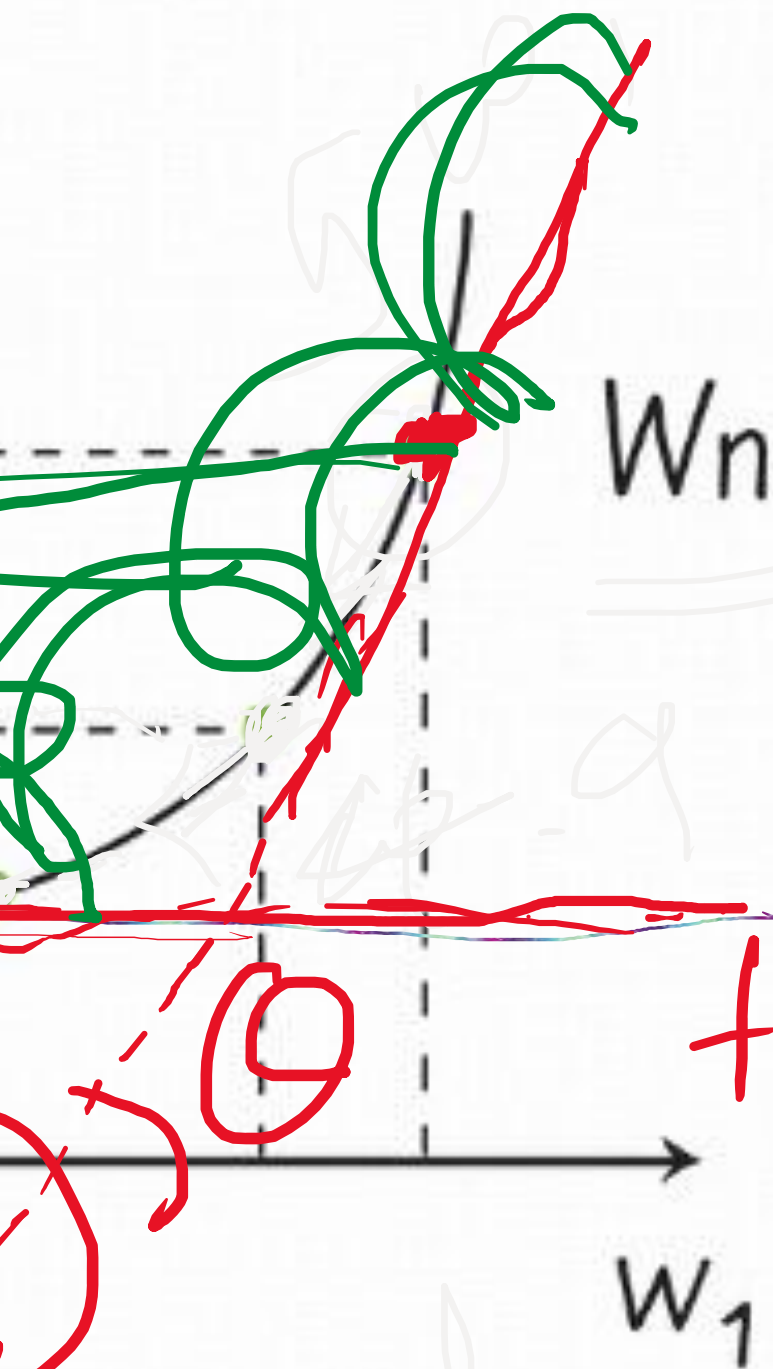


Loss



minimum
loss

Gradient descent



$$W_{\text{new}} = W_{\text{old}} -$$

$$+ \tan \theta = \frac{df}{d\theta}$$

$$\eta = \text{learning rate}$$

ent

$$0.01 * q = h \cdot q$$

$$\eta \left[\frac{\Delta L}{dw} \right]$$

$$\downarrow$$
$$\frac{dy}{dx} =$$

arning rate (0.01 0.01)

$$\frac{d}{d}$$

ω

$$\frac{L}{P} = 0$$

W - OG - 2 - U

2 = 1

= 5 - 9

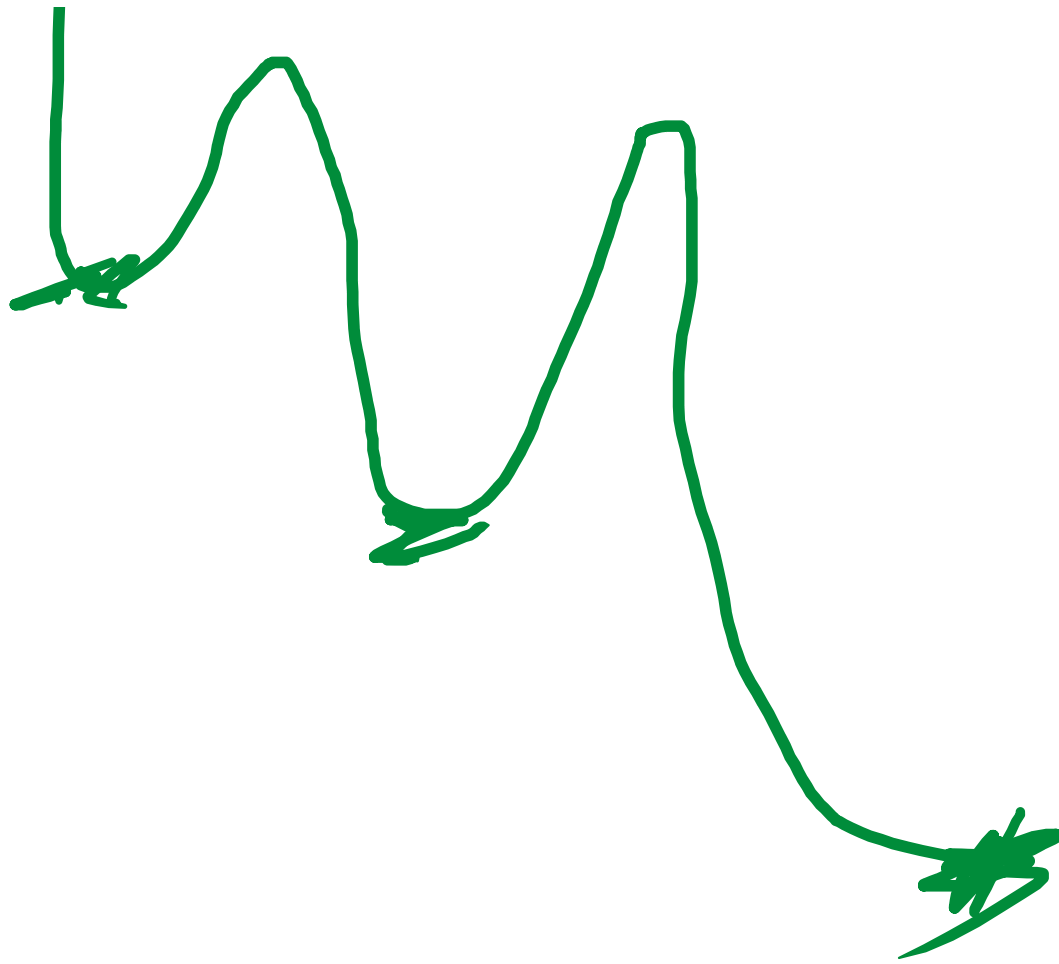
= - 4

$$\frac{V}{d\omega}$$

$$= \omega_0 - \frac{dV}{d\omega}$$

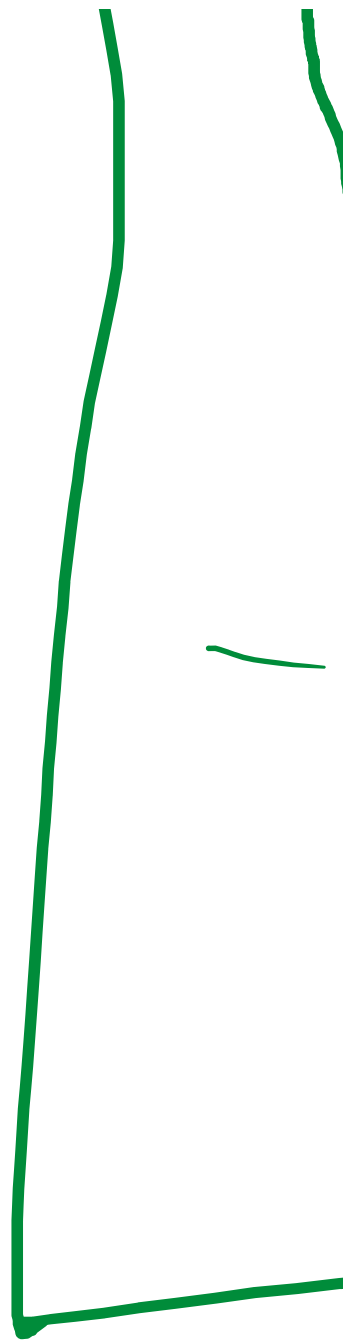
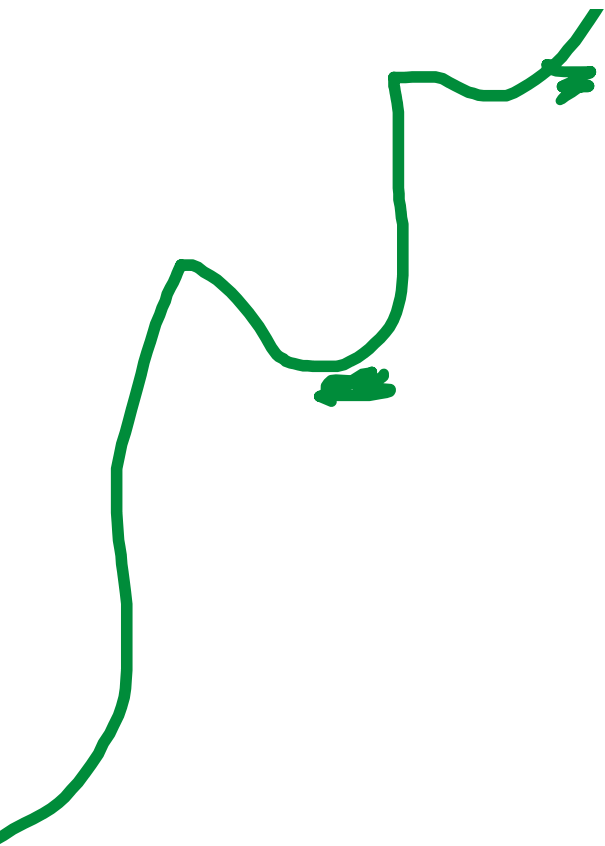


if linear



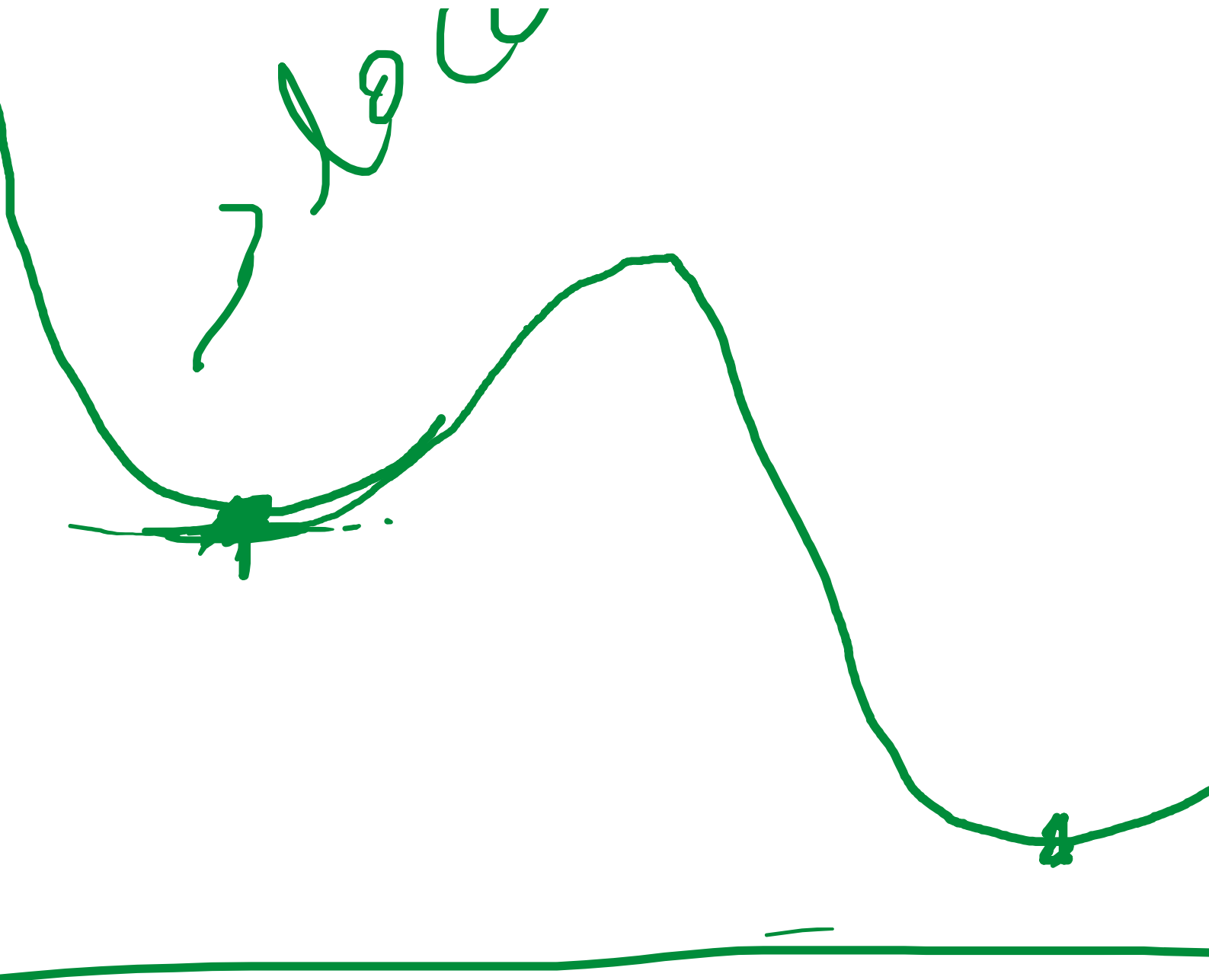
running but

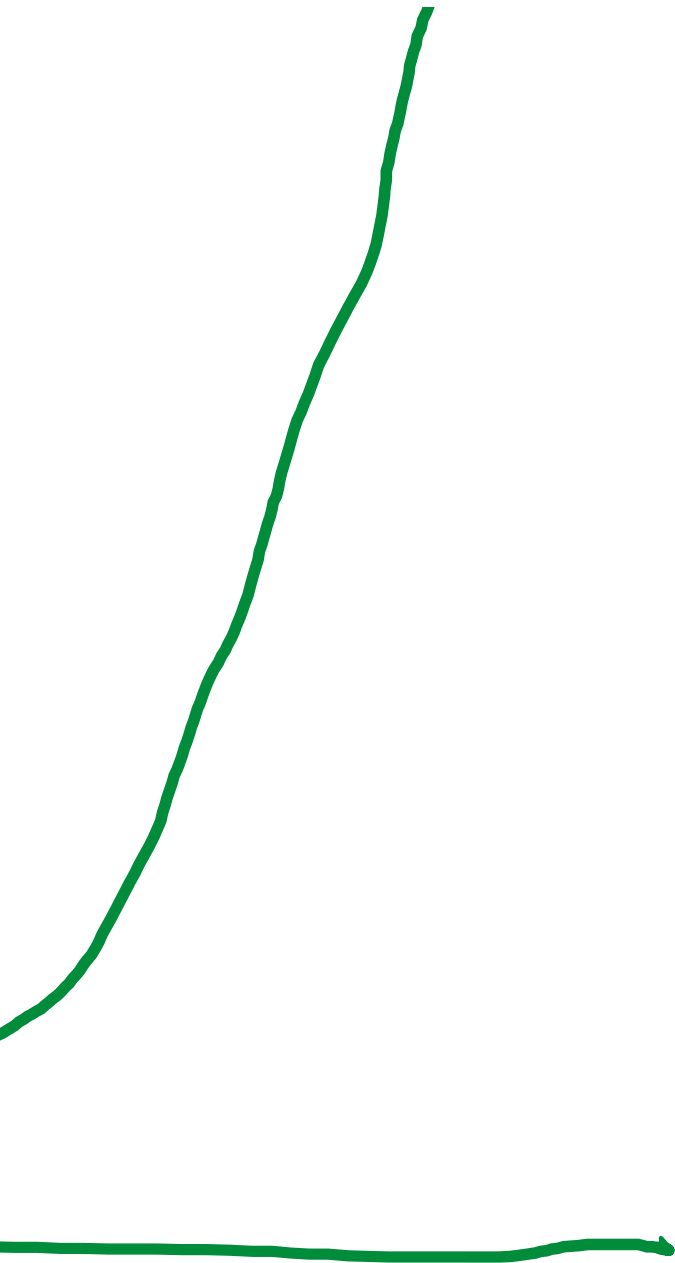
small $\Rightarrow$ we



e

rights are



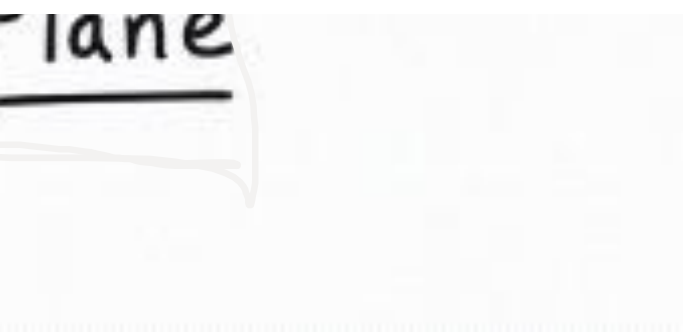


W



Maternal rate $\rightarrow$ ex y $y =$

$$2n_1 + 5n_2 + 7n_3 + 3 \rightarrow \text{Hyper}$$



Once reaching
iteration

g global minima
on stops

$$v_n = w_0 -$$

$$h = 4.91 - 0.1$$

$$= 4.86$$

00

0.01 X 0 ~~0~~

21x5

$$n = \omega_0$$

ω

2

= - 4

n = - 4 -

= ~~9~~ -

5

7

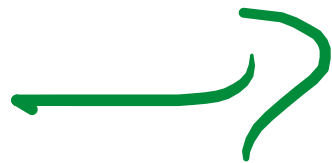
5 mar →

update

\$10 i

ing

ing



easy

o

depth

FO

gm

